ESSENTIAL MATHEMATICS GRADE 10: STATISTICS 1

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 3.0: Statistics and Probability
Sub-Strand	Sub-Strand 3.1: Statistics 1
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Ungrouped data: collecting and organising raw data from real-life contexts – Frequency distribution tables for discrete (ungrouped) data – Measures of central tendency: mean of ungrouped data – Measures of central tendency: mode of ungrouped data – Measures of central tendency: median of ungrouped data – Representing data using bar graphs – Representing data using line graphs – Representing data using pie charts – Reading, extracting and interpreting information from bar graphs, line graphs and pie charts – Using digital devices to draw graphs and charts – Data-driven decision making: drawing conclusions from organised data
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) draw a frequency distribution table for ungrouped data b) determine mean, mode and median of ungrouped data c) represent data using bar graphs, line graphs and pie charts d) interpret data from bar graphs, line graphs and pie charts e) appreciate the importance of data organisation and interpretation in making effective decisions
Core Competencies	– Critical Thinking and Problem Solving: the learner develops interpretation and inference skills when collecting data, organising it into frequency distribution tables, computing measures of central tendency, and drawing conclusions from graphs and charts about real situations in the school or Kenyan community. – Digital Literacy: the learner develops the skills of connecting and interacting with digital technology when using digital devices (learner digital devices, mobile phones, spreadsheet tools) to draw bar graphs, line graphs and pie charts to represent collected data. – Communication and Collaboration: the learner collaborates with peers when collecting community data, sharing findings, extracting information from graphs and presenting data-driven conclusions to classmates. – Learning to Learn: the learner acquires new knowledge independently and in groups as they explore different graph types, compare their representations and reflect on which visual best communicates a given data set.

Core Values	– Integrity: the learner displays honesty while working in groups, collecting data and constructing frequency distribution tables — recording only accurate, unaltered observations. – Love: the learner is considerate of others when working with peers to extract information from graphs and charts and draw conclusions, listening respectfully to different interpretations. – Responsibility: the learner takes ownership of the data collection project, ensuring data from the school or community is gathered ethically and stored carefully on digital devices. – Unity: the learner demonstrates fairness and team spirit when sharing tasks during group data collection, graph construction and presentation activities.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners formulate a community or school question (e.g., "Which means of transport do most learners in our school use?") that drives the entire data collection exercise, mirroring the scientific practice of problem definition. – Planning and Carrying Out Investigations: learners design a simple data-collection plan — deciding what data to collect, from whom, using tally sheets — before gathering real ungrouped data from peers or the school environment. – Analysing and Interpreting Data: learners organise raw data into frequency distribution tables, compute mean, mode and median, and construct bar graphs, line graphs and pie charts to reveal patterns and support claims. – Constructing Explanations and Designing Solutions: learners use interpreted graphs and computed averages to make evidence-based recommendations to the school or community (e.g., advising the school administration on transport planning or canteen menu choices). – Obtaining, Evaluating and Communicating Information: learners present their data project findings to peers using charts drawn on digital devices or graph paper, evaluate the clarity of different representations and communicate data-backed conclusions.
Pertinent & Contemporary Issues (PCIs)	– Life Skills — Decision Making: the learner makes informed decisions when identifying patterns from graphs and charts and drawing conclusions about real school or community situations (e.g., most popular meal in the school canteen, dominant tree species in the school compound). – Education for Sustainable Development: the learner collects and analyses environmental or community data (e.g., waste types generated in school, water usage patterns) and uses statistical findings to recommend sustainable practices. – Social Cohesion: the learner works harmoniously with peers from different backgrounds during group data collection and presentation tasks, appreciating diverse perspectives in interpreting the same data set.
Career Connections	– Journalism and Media: statistical literacy — reading and interpreting charts and graphs — is fundamental for journalists reporting on Kenyan election results, health surveys or economic trends in outlets such as Nation Media or KBC. – Hospitality Industry: managers of hotels and restaurants in Nairobi, Mombasa or Kisumu use frequency distributions and averages to analyse guest feedback, plan menus (e.g., most ordered dishes like nyama choma or pilau) and forecast occupancy. – Sports Management and Coaching: coaches and analysts use measures of central tendency and graphical data to evaluate athlete performance, track training progress and make selection decisions for Kenyan athletics teams. – Guidance and Counselling: school counsellors use data on student welfare, attendance and performance (organised into frequency tables and charts) to identify learners who need support and to make evidence-based recommendations. – Business and Sole Proprietorship: small-business owners in Kenyan markets use bar graphs and averages to track daily sales of items such as mandazi, maize or beans, identify peak trading days and make restocking decisions. – Agriculture: farmers and extension officers in the Rift Valley or Kericho use line graphs and frequency data to monitor rainfall patterns, crop yields and market prices, enabling data-informed planting and selling decisions.
Focus for Lessons	This unit introduces Grade 10 Essential Mathematics learners to the foundational concepts of statistics through the lens of real Kenyan community and school data. Learners progress from collecting and organising raw ungrouped data into frequency

	distribution tables, through computing measures of central tendency (mean, mode and median), to constructing and critically interpreting bar graphs, line graphs and pie charts. The unit is anchored in a community data investigation project in which learners collect genuine data from their school environment — such as modes of transport used to school, favourite foods in the canteen, or tree species in the compound — and use statistical tools to make evidence-based recommendations, developing both mathematical competency and the habit of data-driven decision making.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems? KICD KEY INQUIRY QUESTION: 1. How do we use statistics in day-to-day life?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Canteen Queue Puzzle at Mwangaza Secondary School" Every lunch break at a Kenyan secondary school, the canteen sells several different meals — ugali with sukuma wiki, chapati, githeri, rice and beans, and mandazi. The canteen manager, Mama Wanjiku, notices that on some days she runs out of ugali and sukuma wiki before all learners are served, while on other days she is left with large amounts of unsold githeri. She loses money on unsold food and frustrates hungry learners when popular meals run out early. She has no written records of what learners buy each day — she just guesses from memory. A group of Grade 10 learners decides to help Mama Wanjiku. They collect one week's lunch-order data from 120 learners in their school. When they lay out the raw data — a long, unorganised list of meal choices — it looks like meaningless noise. But when they begin to organise the numbers into a frequency distribution table and then draw a bar chart, a clear pattern emerges: ugali with sukuma wiki is chosen by nearly half the learners every single day, while githeri is the least popular option. The mean number of ugali portions sold per day is 58, the mode is 60 and the median is 57. The pie chart shows that ugali accounts for 47% of all orders. What is puzzling: Why did Mama Wanjiku's memory-based guesses fail so badly, even though she serves learners every day? How can a simple table of numbers and a graph reveal patterns that a human mind misses? And can the same statistical tools that solved Mama Wanjiku's problem be used to answer bigger questions — about transport, health, or the environment — in the wider Kenyan community? This phenomenon is puzzling because it shows that raw, unorganised data is essentially invisible information. Only after systematic organisation, computation and visual representation does the data become a decision-making tool. Learners are motivated to understand WHY and HOW statistical methods transform noise into knowledge.
Storyline Thread	Lesson 1 — "What Is the Data Telling Us?" (Predict + Phenomenon Launch) Learners are shown Mama Wanjiku's unorganised list of 120 meal choices (raw data on the board). They predict — before any instruction — which meal is most popular, writing individual estimates. Cold-call 5 learners to share predictions and reasoning (wait time: 30 seconds). Introduce the anchoring phenomenon formally. Learners brainstorm in pairs: what information would help Mama Wanjiku? Introduce the concept of data collection; learners design a simple tally sheet to collect ungrouped data from classmates (favourite school meal, mode of transport to school, or preferred sport). By end of lesson each group has a raw data set of at least 30 entries. Connect back: "Our raw list looks just like Mama Wanjiku's — what do we do next?"

	Lesson 2 — "Organising the Noise: Frequency Distribution Tables" (Observe + Explain) Learners receive their raw data sets from Lesson 1. Teacher models construction of a frequency distribution table using the transport tally supporting phenomenon (school-gate data). Learners construct frequency distribution tables for their own data sets in groups. Teacher circulates — ask 3 targeted questions: "What is the highest frequency? What does that tell us about Mama Wanjiku's problem? Could she have seen this pattern without the table?" Learners compare tables across groups and notice that organisation reveals patterns invisible in raw data. Connect: update the Driving Question Board (DQB) — "A frequency table organises data, but can it tell us the 'typical' value? What number best represents the whole data set?" Lesson 3 — "Finding the Middle Ground: Mean and Mode" (Explain + Model Building) Using the canteen data phenomenon, learners compute the MEAN number of ugali portions per day (guided example) and the MODE of meal choices. Teacher uses think-pair-share (wait time: 45 seconds per prompt): "Why might the mean and mode give different answers? Which would Mama Wanjiku find more useful?" Learners compute mean and mode for their own group data sets. Connect to the Kericho rainfall supporting phenomenon — what does the mean monthly rainfall tell a tea farmer? Learners add mean and mode to their cumulative data model. DQB update: "We have mean and mode — is there a third measure of 'typical'?" Lesson 4 — "The Middle Value: Median" (Explain + Model Building) Introduce median using a simple ordering activity — 9 learners stand in a line ordered by number of siblings; the middle learner's value is the median (kinaesthetic engagement). Learners order their own data sets and identify the median. Teacher cold-calls 4 learners to explain why ordering matters for median but not for mode (wait time: 30 seconds). Compare mean, mode and median for the canteen data: mean = 58, mode = 60, median = 57 ugali portions. Ask: "If Mama Wanjiku uses the mode to plan, will she over-order or under-order?" Learners update their group data model with all three measures of central tendency. DQB update: "We can now describe data with numbers — but how do we show it visually so anyone can understand it at a glance?" Lesson 5 — "Seeing the Data: Bar Graphs and Line Graphs" (Observe + Explain) Introduce bar graphs using the Nairobi County health survey supporting phenomenon (pre-printed bar chart). Learners read and interpret the chart — teacher asks: "Which sub-county needs the most malaria resources? How does the bar graph make this obvious?" Learners construct bar graphs for their own group data sets on graph paper. Introduce line graphs using the Kericho rainfall supporting phenomenon — learners trace trends across months. Discuss: "When is a line graph better than a bar graph?" (continuous vs. discrete data). Learners draw a line graph for a provided monthly sales data set from a Kisumu market vendor. Connect to phenomenon: "If Mama Wanjiku tracked daily ugali sales over a month as a line graph, what would she see?" DQB update: "Bar graphs and line graphs show frequency and trend — but how do we show proportions (parts of a whole)?" Lesson 6 — "Slicing the Circle: Pie Charts" (Explain + Model Building) Use the KNBS mobile-phone usage pie chart supporting phenomenon as the entry point. Learners observe and interpret the chart before any instruction (predict phase within lesson). Teach the method for calculating sector angles (frequency ÷ total × 360°) using the canteen data. Learners construct pie charts for their own data sets using protractors and compasses. Teacher circulates — cold-call 3 learners: "What percentage of your data does the largest slice represent? Does this match your frequency table?" Learners use a digital device (LDD or mobile phone) to recreate their pie chart using a spreadsheet tool and compare with hand-drawn version. Connect: "Mama Wanjiku could show this pie chart to the school principal to justify buying more ugali ingredients." DQB update: "We can now draw three types of graphs — but which graph is best for which type of data?" Lesson 7 — "Reading Between the Lines: Interpreting Graphs and Charts" (Explain + DQB Consolidation) Provide learners with three unfamiliar graphs (bar graph of KCSE subject enrolment in a Nairobi school; line graph of Lake Victoria

	water levels 2015–2024; pie chart of land use in Rift Valley County). In groups, learners extract specific information, identify trends, and make two evidence-based recommendations per graph. Groups present to class; peers use a structured feedback protocol ("One thing that is clear… One question I still have…"). Teacher facilitates whole-class DQB update: compile all questions learners still have, categorise answered vs. unanswered questions. Connect back to phenomenon: "Could we now build a complete statistical report for Mama Wanjiku using all our tools — table, mean/mode/median, bar graph, line graph, pie chart?" Lesson 8 — "Helping Mama Wanjiku: Data Project Presentation and Model Finalisation" (Model Building + Project + Assessment) Learners finalise their group data project (begun in Lesson 1): organise collected data into a frequency distribution table, compute mean, mode and median, construct at least two graph types (one on graph paper, one on a digital device), write two data-driven recommendations for the school or community. Groups present their completed statistical reports (3 minutes per group). Peers and teacher assess using the suggested rubric (accuracy of frequency table, correctness of mean/mode/median, quality of graphs, validity of conclusions). Final DQB review: each learner writes one sentence answering the driving question — "How can statistics help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?" Teacher collects exit cards. Connect to careers: brief discussion on how journalists, farmers, health workers and sports coaches in Kenya use exactly these tools every day.
Supporting Phenomena	– Transport Tally at the School Gate (Lesson 1–2): A teacher stands at the school gate for one morning and tallies how learners arrive — matatu, boda boda, walking, cycling, or private car. The resulting raw tally of 200 entries looks chaotic, but when converted into a frequency distribution table the dominant mode of transport (walking) becomes immediately obvious — sparking curiosity about why organising data changes what we can see. – Rainfall Records at Kericho Tea Estates (Lesson 4–5): A printed line graph from a Kericho tea estate shows monthly rainfall totals over one year. Learners notice that tea farmers can predict when irrigation is needed simply by reading the graph — illustrating how line graphs reveal trends over time in a way that a table of numbers alone cannot. – Nairobi County Health Survey Bar Chart (Lesson 5–6): A bar chart from a Nairobi County public health report shows the number of children under five treated for malnutrition, malaria, respiratory infections and diarrhoea in different sub-counties. Learners observe that the chart instantly shows which disease is most prevalent — demonstrating how bar graphs support resource-allocation decisions in real government contexts. – Mobile Phone Usage Pie Chart (Lesson 6–7): A pie chart from the Kenya National Bureau of Statistics (KNBS) showing the proportion of Kenyans who use mobile phones for calls only, mobile money (M-Pesa), internet browsing, and social media. Learners are puzzled that a simple circle divided into slices can communicate proportional relationships more powerfully than a list of percentages — motivating deep engagement with pie chart construction and interpretation.

LESSON 1 (40 minutes): What Is the Data Telling Us? — Launching the Canteen Queue Puzzle	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and activate prior knowledge about data, so that learners feel genuinely puzzled by why Mama Wanjiku's memory-based guesses fail and are motivated to use statistics to help her.
Knowledge	Learners will know that raw, unorganised data is difficult to interpret, and that statistics provides systematic tools — frequency tables, graphs, and measures of central tendency — for organising and making sense of data.
Skills	Learners will be able to read a raw data list, make a prediction about patterns in meal-choice data, and begin constructing a frequency tally for ungrouped discrete data.
Attitudes	Learners will appreciate that honest, systematic data collection is more reliable than memory or guesswork, and will show integrity when recording observations.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	This is the opening lesson of the Statistics 1 sub-strand. It launches the anchoring phenomenon — 'The Canteen Queue Puzzle at Mwangaza Secondary School' — by placing learners in Mama Wanjiku's shoes. Learners confront a long, unorganised list of 30 raw meal-order entries, try to answer questions from memory, and immediately feel the cognitive frustration of raw data. This productive confusion is the engine that will drive the entire unit. The lesson ends by opening the Driving Question Board (DQB) and establishing the unit's central question.
Safety Notes	No physical hazards. Remind learners to be respectful when discussing food choices and economic circumstances; some learners may be sensitive about meal affordability. Ensure psychological safety so all predictions are treated as valid starting points.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon through a short story about Mama Wanjiku's canteen problem at Mwangaza Secondary School. They examine a printed (or displayed) raw list of 30 meal orders — ugali with sukuma wiki, chapati, githeri, rice and beans, and mandazi — and attempt to answer questions about it from memory alone, mimicking exactly what Mama Wanjiku does. After experiencing firsthand how hard it is to see patterns in raw data, learners make and share explicit predictions about what they think the data shows. They are then guided to begin a simple tally/frequency table, which starts to reveal structure. The lesson closes by co-creating the first version of the Driving Question Board, where learners post their wonderings and predictions to anchor the rest of the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles]()	Learners read or listen to a two-paragraph story about Mama Wanjiku's canteen problem. They are then shown (projected or on a printed handout) a raw list of 30	1. Read aloud the Mama Wanjiku story with expression: 'Every lunch break, Mama Wanjiku guesses how much food to cook — just from memory. Some days she runs	Think-Pair-Share on raw data: By asking learners to predict without counting, the teacher surfaces prior intuitions and creates cognitive dissonance — learners	Observable indicator: Every learner has written at least one prediction on their slip. Teacher circulates during individual writing (2 min) and notes whether learners

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	meal-order entries in random order — e.g., 'ugali, chapati, githeri, ugali, rice and beans, ugali, mandazi, ugali, chapati, ugali, githeri, ugali, rice and beans, chapati, ugali, mandazi, ugali, githeri, ugali, chapati, rice and beans, ugali, ugali, mandazi, githeri, chapati, ugali, rice and beans, ugali, ugali'. Without any tallying or counting, learners individually write answers to three questions on a small slip of paper: (1) Which meal is ordered most? (2) Which is ordered least? (3) How many times is ugali ordered? They then share predictions with a partner using a Think-Pair-Share.	out of ugali. Other days she is left with pots of cold, unsold githeri. Today, we are going to find out WHY her guesses fail.' [WAIT 5 seconds.] 2. Display the raw list. Say: 'This is ONE day of meal orders — just 30 learners. Mama Wanjiku has 120 learners every day. Look at this list for 30 seconds. Do NOT count. Just look.' [Time 30 seconds of silence.] 3. Distribute prediction slips. Say: 'Without counting, write your best guess for each question. There is no wrong answer right now — we are predicting.' [WAIT TIME: allow 2 minutes of silent individual writing.] 4. Prompt Think-Pair-Share: 'Turn to your partner. Compare your predictions. Did you agree? Why or why not? You have 90 seconds.' 5. Cold-call 4 pairs (select mix of confident and quieter learners): 'Aisha and Brian — what did you predict for the most popular meal, and why?' After each response, say: 'Interesting — hold that thought.' Do NOT confirm or deny any prediction yet. 6. Write the range of predictions on the board under the heading: 'OUR PREDICTIONS — Before We Count.'	will disagree with each other, immediately showing that unaided memory/guessing is unreliable. This mirrors Mama Wanjiku's real problem and creates the motivational hook for statistical tools.	are attempting all three questions. Collects 4–5 slips to scan the range of guesses. Looks for: are learners' predictions spread widely (shows raw data is genuinely ambiguous) or are they consistent (may indicate some learners tried to count — ask them to explain their strategy).
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing	Learners now receive a larger printed version of the same 30-item raw list (or view it on screen). In groups of three, they are given 5 minutes to count however they like — no structure is imposed yet. Each group records their count on a mini-whiteboard or piece of paper. Groups will almost certainly get different totals for ugali, revealing that even when people try to count raw data they make errors. The class then compares group totals projected on the	1. Distribute the raw list printout (A5 card). Say: 'Now you MAY count. In your group of three, count how many times each meal appears. You have 5 minutes. Use any method you like.' [Set a visible 5-minute timer.] 2. While groups work, circulate. Ask probing questions without giving answers: 'How are you keeping track? What happens when you lose your place?' Note which groups use tally marks spontaneously — these groups will be referenced later. 3.	Structured controversy through group counting: Deliberately allowing unstructured counting guarantees different totals across groups. This observable disagreement is not a failure — it IS the lesson. It provides concrete, lived evidence that raw data without an organisational system produces unreliable information. The teacher names this: 'What you just experienced is why statisticians invented the frequency distribution table.'	Observable indicator: Each group produces a written count for all five meal types. Teacher notes (a) which groups attempted a systematic method independently (evidence of prior knowledge to build on), and (b) how widely the ugali counts vary across groups (wider spread = stronger evidence of the need for a tally system). If all groups agree, teacher asks: 'How confident are you — would you bet Mama Wanjiku's income on this count?'

Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	board.	After 5 minutes, call the class back. Say: 'Hold up your ugali count — on the count of three.' [Scan boards.] Write all different answers on the board: e.g., 14, 15, 16, 17. Ask: 'We all looked at the SAME list. Why do we have DIFFERENT answers?' [WAIT TIME: 15 seconds of silence before cold-calling.] 4. Cold-call 3 learners: 'Kamau — what do you think caused the difference?' Accept all answers. 5. Ask: 'What would make counting easier and less error-prone?' [WAIT 10 seconds.] Accept suggestions. If no one mentions a tally or table, say: 'One group over here used a system — can you show us what you did?' [Invite that group to share.] 6. Bridge to phenomenon: 'This is EXACTLY Mama Wanjiku's problem — but she has 120 orders, not 30, and she does this from MEMORY, without even writing anything down.'		
Explain Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the tally/frequency distribution table as the solution to the counting problem just experienced. Learners reconstruct the 30-item count using a class tally table drawn on the board, with individual learners calling out items one by one. The class arrives at an agreed, verified frequency table. Learners are then introduced to the terms: raw data, frequency, frequency distribution table, tally marks. They connect each term back to Mama Wanjiku: 'This table is the tool that could have saved Mama Wanjiku from running out of ugali.'	1. Draw a blank 3-column table on the board: Meal | Tally | Frequency. Say: 'We are going to build this table TOGETHER, one order at a time. I will point to each item on the list and YOU call it out.' 2. Point to item 1 on the projected list. Wait for the class to call the meal name aloud. Make the tally mark. Continue through all 30 items, pausing every 5 marks to do the bundle: 'Four strokes, then the diagonal — that gives us a bundle of five. Why do we bundle? Because bundles are fast to count.' 3. When the tally is complete, say: 'Count your bundles and singles for each meal and give me the frequency.' Cold-call one learner per row (5 cold-calls total). Record frequencies. 4.	Whole-class co-construction of a frequency table: By building the table together — with every learner seeing the transformation from raw list to organised table in real time — the teacher makes the statistical process transparent and memorable. Connecting every step back to Mama Wanjiku keeps the purpose authentic. The vocabulary introduction at the END (after experience) follows an inquiry-first approach: learners understand the NEED for the word before they learn it.	Observable indicator: When the teacher cold-calls for each row's frequency, every called learner can state the correct number after counting tally marks. Teacher checks that the class total equals 30. Secondary check: ask 2–3 learners to explain IN THEIR OWN WORDS what 'frequency' means in the context of Mama Wanjiku's meals — listening for: 'how many times each meal was ordered' or equivalent. If learners cannot explain, reteach using: 'Frequency just means — how many? How many ugali? How many githeri?'

		Ask the class: 'Add up all frequencies. What should the total be?' [WAIT 10 seconds.] Verify total = 30. Say: 'This is a CHECK — if your frequencies don't add up to your total data points, you have made an error.' 5. Now compare to prediction slips: 'Look at your prediction from Phase 1. Were you right about ugali being most popular? Why was it hard to tell from the raw list?' [WAIT 15 seconds.] Cold-call 3 learners. 6. Introduce vocabulary explicitly: write on board — RAW DATA (the unorganised list), TALLY (a mark for each item), FREQUENCY (the count), FREQUENCY DISTRIBUTION TABLE (organising data into a table). Ask: 'Which of these four things does Mama Wanjiku currently have? Which does she lack?' [WAIT 10 seconds; cold-call 2 learners.] 7. Say: 'The frequency distribution table has turned Mama Wanjiku's invisible data into visible information. But we can do even more — over the next lessons we will calculate averages and draw graphs that make the pattern impossible to miss.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Summary Statistics,]	Learners are introduced to the unit Driving Question: 'How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?' Each learner writes one question or wondering on a sticky note (or folded paper strip). Questions are grouped on the DQB under two headings: 'Things we are wondering about DATA' and 'Things we are wondering about GRAPHS and NUMBERS'.	1. Reveal the unit Driving Question on the board (written in large text). Read it aloud twice. Say: 'This is the big question we will answer together over the next lessons. Mama Wanjiku is waiting for our help — but so are many other Kenyans. Think about: what do you WANT to know? What are you still puzzled about?' [WAIT 15 seconds of silent thinking.] 2. Distribute sticky notes or paper strips. Say: 'Write ONE question — something you are genuinely wondering about. It can be about	Driving Question Board launch: The DQB externalises and values learner curiosity, making it a living artefact of the class's growing understanding. By having learners write questions AFTER the Observe and Explain phases, the questions are richer and more specific than if asked cold. Grouping questions under 'DATA' vs 'GRAPHS and NUMBERS' previews the sub-strand's two major threads and helps learners see the trajectory of the unit.	Observable indicator: Every learner has posted at least one question on the DQB. Teacher reads all questions before the next lesson and categorises them to inform planning. Key diagnostic signal: if many learners ask 'what is mean/mode/median?' this confirms they need Lesson 2. If learners ask more sophisticated questions ('can we compare two schools?', 'what if data changes each week?'), note these for extension tasks. If any learner's question reveals a misconception

Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners do a gallery-walk past the board and place a small tick on any question they share.	data, about graphs, about numbers, or about Mama Wanjiku. You have 2 minutes.' [Silent individual writing.] 3. Invite learners to come up and place their notes on the board. Read 4–5 aloud, paraphrasing: 'Fatuma is asking — how do we draw a graph from a table? Great — that is coming in Lesson 3. Peter is asking — what is the average number of ugali portions? Excellent — that is Lesson 2.' Do NOT answer the questions now — the DQB is for TRACKING, not immediate resolution. 4. Say: 'This board belongs to you. Every lesson, we will come back to it and move questions to the ANSWERED side as we learn. By the end of this unit, every question on this board should have an answer — and we will use those answers to write a real report for Mama Wanjiku.' 5. Point to the completed frequency table: 'Look at what we already know from just ONE lesson. We can already start answering our predictions. What have we learned so far that is NEW compared to what Mama Wanjiku currently knows?' [WAIT 10 seconds; cold-call 2 learners.]		(e.g., 'isn't the most popular meal always the one that costs least?'), plan to address it explicitly in Lesson 2.
Model Building Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics,	As a closing model-building task, learners individually sketch a simple annotated diagram (a 'data journey map') showing the path from raw data to organised information. The map has three labelled boxes connected by arrows: RAW DATA LIST → FREQUENCY DISTRIBUTION TABLE → (a question mark labelled 'What comes next?'). Learners annotate the first arrow with what they learned today (tally marks, counting, checking totals)	1. Say: 'Before we finish, I want you to draw a simple map — a journey that data takes. Draw three boxes connected by two arrows.' [Demonstrate on board quickly.] 2. Say: 'Label box 1: RAW DATA — the messy list. Label box 3: a question mark. For now, write WHAT COMES NEXT? Label box 2: FREQUENCY TABLE — organised data.' 3. Say: 'On the FIRST arrow, write at least two things we did today to move from raw data to a frequency table.'	Cumulative Data Journey Map (progressive model building): The three-box annotated diagram is the learner's running model of the statistical process. It is deliberately incomplete — the question mark on arrow 2 is a designed gap that creates anticipation. Over the unit, learners will add: measures of central tendency (Lesson 2), bar graphs and line graphs (Lesson 3), pie charts (Lesson 4), and interpretation/decision-making (Lesson 5). The model makes	Observable indicator: Each learner's map correctly labels RAW DATA and FREQUENCY TABLE, and the first arrow contains at least one accurate description of today's process (e.g., 'tally marks', 'count each meal', 'check total = 30'). Teacher reviews maps at the end of the lesson. Red flag: any learner who cannot label or annotate the first arrow needs individual follow-up — they may not have understood the core move from raw data to tally to

Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and leave the second arrow as a question — to be filled in across subsequent lessons as they add graphs and calculations.	[WAIT TIME: 2 minutes silent individual work.] 4. Say: 'On the second arrow — the question mark — write what you THINK might come next. A guess is fine.' [WAIT 1 minute.] 5. Cold-call 3 learners to share their 'what comes next' guesses. Write these on the board. Say: 'These guesses are your model of statistics right now. We will update this map every lesson. By Lesson 5, your map will show the COMPLETE journey from Mama Wanjiku's raw orders to a report that saves her business.' 6. Collect the maps or have learners keep them in their exercise books — they will be revised in every subsequent lesson. 7. Close: 'Today, you did what Mama Wanjiku never did — you organised data. Next lesson, we calculate the mean, mode and median so we can tell her exactly how many portions of each meal to cook. See you then.'	conceptual growth visible to both learner and teacher.	frequency. Green flag: learners who write 'graphs' or 'average' on the question mark arrow are ready for extension in Lesson 2.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did the raw-data counting activity (Phase 2) genuinely produce different totals across groups? If all groups agreed immediately, the raw list may have been too short or too obvious — consider using a longer, less patterned list in future. (2) How rich were the DQB questions? If questions were shallow ('what is a graph?'), the phenomenon story may need more detail or emotional resonance to generate genuine curiosity — consider adding a line about Mama Wanjiku losing money. (3) Were all learners able to complete the frequency table co-construction, or did some struggle to follow the tally? If more than 20% of learners could not confidently state a row's frequency when cold-called, open Lesson 2 with a brief 5-minute independent tally practice before moving to mean and mode. (4) Did the Data Journey Map reveal any concerning misconceptions? Review maps before Lesson 2. (5) Note which learners were reluctant to share predictions — create structured opportunities in Lesson 2 for those learners to contribute through written responses rather than cold-call only.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined a raw list of 30 meal orders from Mama Wanjiku's canteen and experienced firsthand how difficult it is to answer simple questions (which meal is most popular? how many ugali orders?) from unorganised data — even when they were allowed to count freely. Different groups reached different totals for the same list.
What did I learn?	Raw data is essentially invisible information. A frequency distribution table — built using tally marks — organises data so that

	patterns (such as ugali being the most popular meal) become immediately visible and verifiable. Key vocabulary introduced: raw data, tally, frequency, frequency distribution table.
How does this explain the phenomenon?	Mama Wanjiku's memory-based guessing fails because the human mind cannot reliably track and compare frequencies across many transactions without a systematic recording tool. The frequency distribution table is the first step in transforming raw data into a decision-making tool — exactly what Mama Wanjiku needs to know how much of each meal to cook each day.

LESSON 2 (40 minutes): From Chaos to Clarity: Collecting Raw Data to Help Mama Wanjiku	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners develop the foundational data-collection skill of designing and using a tally sheet to gather ungrouped data from peers, directly mirroring the raw-data problem faced by Mama Wanjiku at Mwangaza Secondary School.
Knowledge	Learners will know: (a) what ungrouped (raw) data is and why it looks like 'meaningless noise' before organisation; (b) how to design a simple tally sheet with clearly defined categories; (c) how to accurately record at least 30 data entries using tally marks.
Skills	Learners will be able to: (a) identify a researchable question linked to a real school or community problem; (b) design a tally sheet with appropriate categories; (c) collect ungrouped data systematically from classmates using tally marks; (d) present their raw data set to the class and articulate what the data does NOT yet tell them.
Attitudes	Learners will appreciate that honest, systematic data collection — rather than memory or guesswork — is the first and most critical step in turning raw information into a decision-making tool, and will demonstrate integrity while collecting peer data.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	This is Lesson 2 of 8 in the evidence-gathering sequence. In Lesson 1, learners were introduced to Mama Wanjiku's canteen problem and made intuitive predictions about meal popularity. Today they move from prediction to action: they design their own tally sheets and collect real ungrouped data from classmates — recreating exactly the messy, unorganised list that Mama Wanjiku never had. By the end of the lesson each group holds a raw data set of at least 30 entries and experiences first-hand why that list still cannot answer Mama Wanjiku's question — setting up the need for a frequency distribution table in Lesson 3.
Safety Notes	No physical hazards. Remind learners to respect peers' responses (e.g., meal preferences or transport modes) and not to ridicule or pressure classmates. Data collected remains anonymous — no names are recorded on tally sheets.

B. LESSON OVERVIEW

Learners begin by recalling their Lesson 1 predictions about Mama Wanjiku's most popular meal, then examine a deliberately chaotic 'raw data wall' — a board display of 120 unsorted meal-choice entries — to feel the frustration of unorganised data. Working in groups of 4–5, they brainstorm what information would genuinely help Mama Wanjiku, identify a researchable question of their own (favourite school meal, mode of transport to school, or preferred sport), design a tally sheet, and collect ungrouped data from at least 30 classmates. The lesson closes with groups reading out their raw totals and reflecting on why even 30 entries — unorganised — still cannot answer their research question, anchoring the need for frequency distribution tables in Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean]	Individually, each learner writes on a sticky note or exercise book: (a) which meal they think was most popular in Mama Wanjiku's	Display the raw data wall before learners enter or as the lesson opens. Say: 'This is exactly what Mama Wanjiku's week looked like	Strategy: PREDICT–OBSERVE–EXPLAIN (POE) Launch. Learners articulate a prior belief before any instruction, creating cognitive	Observable indicator: Each learner has a written prediction with a stated reason. Teacher scans the room during the 60-second wait

Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	canteen last week, and (b) one reason WHY they think so. Learners are then shown the 'raw data wall' — 120 meal-choice entries written in random order on the board or a large chart paper (e.g., 'ugali, chapati, githeri, ugali, rice & beans, ugali, mandazi, chapati, ugali …'). Learners study the wall silently for 60 seconds and write a second prediction: 'Can you tell which meal is most popular just by looking at this list? Yes / No — explain in one sentence.'	— 120 meal choices, written down just as they happened. Stare at it for 60 seconds. Write your answer to this question: Can you tell which meal is most popular?' [WAIT 60 seconds — do not speak.] Then cold-call EXACTLY 5 learners using name cards or a random selector. For each, ask: 'What did you predict, and what was your reason?' [WAIT TIME: 10 seconds after each answer before responding.] After the 5th learner, say: 'Notice — some of us guessed ugali, some said chapati. We all used memory and intuition, just like Mama Wanjiku. Today we find out whether there is a better way.'	dissonance when the raw data wall proves unreadable. This dissonance motivates the need for data organisation tools introduced later in the sequence.	and notes whether learners are writing or staring blankly. Cold-call responses reveal whether learners can articulate the difference between a guess and evidence — a baseline for the lesson.
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4–5 learners receive a printed or hand-drawn 'Mama Wanjiku raw data strip' — a strip of paper listing 30 of the 120 meal entries in random order (e.g., 'ugali, chapati, ugali, githeri, ugali, rice & beans, mandazi, ugali, chapati …'). Groups are given 5 minutes to: (a) count how many times 'ugali' appears on their strip using any method they choose (fingers, crossing out, circling), (b) record their count, and (c) note how difficult or easy the counting was. Groups then share their ugali counts aloud. The teacher records each group's count on the board. Counts will likely vary slightly due to miscounting — the teacher deliberately highlights this inconsistency without correcting it yet.	Distribute the data strips. Say: 'Your strip is 30 of Mama Wanjiku's orders. Your only task: count how many times ugali appears. You have 5 minutes — go.' [WAIT — circulate, observe methods used: circling, tallying, pointing, re-reading.] After 5 minutes, call each group to report. Write counts on the board in a column. Say: 'Interesting — Group 1 says 13, Group 2 says 12, Group 3 says 14. Same strip, different answers. Why might that happen?' Cold-call 3 learners. [WAIT TIME: 10 seconds each.] Then ask: 'What tool could have made counting faster AND more accurate?' [WAIT TIME: 15 seconds — expect 'tally marks' from some learners who recall primary school.] Affirm: 'Yes — a tally sheet. That is exactly what we are going to build today.'	Strategy: PRODUCTIVE STRUGGLE with raw data. Learners experience first-hand the unreliability of counting from an unorganised list. The variation in group counts creates authentic motivation for a systematic recording tool (the tally sheet), making the upcoming design task feel necessary rather than arbitrary.	Observable indicator: Groups can produce a count (even if slightly inaccurate) within 5 minutes. Teacher notes which groups used spontaneous tally marks versus re-reading the strip — groups using tallies organically are already demonstrating prior knowledge to leverage. Discrepant counts on the board serve as a whole-class formative data point.
Explain Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean	The teacher introduces the term 'ungrouped data' and explains what a tally sheet is, using a worked example on the board with	Write on the board: UNGROUPED DATA = raw data that has NOT yet been sorted or counted into groups. Say: 'Before Mama	Strategy: WORKED EXAMPLE → GUIDED DESIGN → INDEPENDENT APPLICATION (I Do – We Do – You Do). The	Observable indicator: Each group's tally sheet has a clear research question written at the top, 4–6 mutually exclusive categories, and

Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	a familiar Kenyan context (modes of transport: matatu, boda boda, walking, cycling, school bus). Learners then work in their groups to design their OWN tally sheet for one of three approved research questions: (A) 'What is the favourite school meal among learners in our class?' (B) 'How do learners in our school travel to school each day?' (C) 'What is the most popular sport played by learners in our school?' Each group selects their question, agrees on 4–6 categories, and rules for the tally sheet (e.g., one tally per learner, no names recorded). Groups draft their tally sheet in their exercise books. Teacher reviews and approves each group's sheet before data collection begins.	Wanjiku can make a bar chart or calculate a mean, she needs this — a tally sheet.' Model a tally sheet on the board: draw a two-column table (Category | Tally), fill in 'matatu | IIII II' live, explaining each mark. Then say: 'Now your group will design your own. Choose ONE of these three questions.' [Display the three questions on the board.] 'You have 6 minutes to agree on your question, write your categories, and draw your tally sheet. I will come and stamp-approve it before you start collecting.' [Circulate — check that: categories are mutually exclusive, there are 4–6 options max, 'other' is included as a catch-all.] For groups who finish early, prompt: 'How will you make sure every classmate is counted only once?' After approval, say: 'Now — collect data from at least 30 classmates. Divide the work: two people ask, two people tally. You have 10 minutes. Go.'	teacher models with a Kenyan transport example, making the structure of a tally sheet explicit. Groups then design their own sheet, giving them ownership while the teacher's approval checkpoint prevents procedural errors from corrupting the data collection phase.	a tally column. Teacher uses a physical 'stamp approved' or initialled mark on each sheet before groups begin collecting. Groups that struggle to define categories reveal a misconception about data categories that the teacher addresses immediately with a targeted question: 'Can one learner belong to two of your categories at the same time?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Data collection ends. Each group counts their total tally marks and confirms they have at least 30 entries. Groups then write THREE sticky notes (or index cards) to add to the class Driving Question Board: (1) BLUE — 'Something we now KNOW about data collection' (e.g., 'Tally marks make counting faster and more accurate'). (2) YELLOW — 'Something we are still WONDERING' (e.g., 'How do we know if our data is representative of the whole school?'). (3) PINK — 'A connection to Mama Wanjiku's problem' (e.g., 'Mama Wanjiku's problem is that she never made a tally sheet — she just trusted memory'). Groups post their cards	Signal end of data collection: 'Pens down — count your tallies and write your total at the bottom of your sheet.' Give groups 2 minutes to count. Then distribute sticky notes or index cards. Say: 'Three cards, three colours: BLUE — what you NOW know; YELLOW — what you are still wondering; PINK — how this connects to Mama Wanjiku. You have 3 minutes.' [Circulate — prompt groups who are stuck with: 'Think about what was hard about counting from the raw data strip. What would have helped Mama Wanjiku?'] Once cards are posted, read 2 YELLOW 'wondering' cards aloud — cold-call the group to explain their wondering. Say: 'These wondering	Strategy: DRIVING QUESTION BOARD as a living knowledge-tracking tool. The three-colour card system externalises the difference between established knowledge, open questions, and cross-cutting connections. By physically posting on the DQB, learners see their understanding accumulating across lessons and can return to unanswered questions in later lessons.	Observable indicator: Each group posts all three cards with substantive content (not blank or one-word responses). Teacher reads PINK cards to check whether groups are making the phenomenon connection explicitly. A group that writes 'Mama Wanjiku should use tally sheets' demonstrates understanding of the lesson's core idea; a group that writes only 'we collected data' is not yet connecting to the phenomenon and needs a follow-up prompt.

	under the Driving Question: 'How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?'	questions are exactly what the next lessons will answer. Hold on to your raw data — we will need it tomorrow.'		
Model Building Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Ungrouped Data to Find the Mean (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group presents their raw data tally sheet to the class in a 60-second 'data report': they state their research question, their categories, their total number of responses, and their largest tally count. After all groups report, the teacher returns to the raw data wall from Phase 1. Learners individually complete a 'model update' in their exercise books: they draw a two-column table titled 'What raw data looks like vs. what we need', filling in the left column ('What raw data looks like': unorganised, hard to count, easy to miscount, no clear pattern) and the right column ('What we still need': a way to organise the counts, a way to see which category is biggest, a way to share findings with Mama Wanjiku). The lesson closes with the teacher posing the bridge question for Lesson 3.	Signal presentations: 'Each group has 60 seconds — research question, categories, total responses, biggest category. Group 1, go.' [Use a visible timer. Do NOT allow over-run — this models the discipline of concise data reporting.] After all groups, point to the raw data wall: 'Look at this again. At the start of the lesson, this was unreadable. Your tally sheet is an improvement — but is it enough for Mama Wanjiku to order the right amount of ugali tomorrow?' [WAIT TIME: 15 seconds — cold-call 2 learners.] Then say: 'Draw this table in your books.' [Draw the two-column model on the board and give learners 3 minutes to complete it.] Close with: 'Your tally sheet is step one. Mama Wanjiku has her counts — but she still cannot see a pattern, cannot calculate a mean, and cannot draw a graph. What must she do NEXT with these counts? Think about that tonight. Next lesson, we transform this raw tally into a frequency distribution table.'	Strategy: CUMULATIVE MODEL REVISION — Two-Column Contrast Table. Rather than asking learners to summarise abstractly, the two-column table forces explicit comparison between the current state of their knowledge (raw tallies) and the next need (organised table/graph). This builds a mental model of the full statistical process — data collection → organisation → representation → interpretation — which will be revisited in every subsequent lesson.	Observable indicator: Learners' two-column tables contain at least 2 entries per column with specific, accurate language (e.g., 'hard to count' not just 'messy'; 'frequency distribution table' if the term has been previewed, or 'organised count table'). Teacher collects 3–4 books at random to check before Lesson 3 and uses findings to decide whether a 5-minute review of raw-data limitations is needed at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did all groups reach at least 30 data entries — if not, what slowed them down and how will you adjust time allocation in Lesson 3? (2) Were the research questions learners chose genuinely connected to their school or community reality, or did groups default to abstract topics? If disconnected, consider assigning the transport-to-school question to all groups in Lesson 3 so the whole class shares one data set for comparison. (3) Which groups' YELLOW 'wondering' cards revealed the deepest statistical thinking? Plan to cold-call these groups first in Lesson 3 to activate prior curiosity. (4) Did any learner struggle with the concept of mutually exclusive categories (e.g., listing 'ugali' and 'ugali with sukuma wiki' as separate categories)? If so, this is a priority misconception to

address before frequency distribution tables are introduced. (5) How well did the raw data wall create genuine cognitive dissonance — did learners actually FEEL the frustration of unorganised data, or did it feel like a teacher-contrived activity? If the latter, consider having learners themselves write the 120 entries on the board next time for stronger embodied engagement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined a 'raw data wall' of 120 unsorted meal choices from Mama Wanjiku's canteen and attempted to count ugali entries from a 30-item data strip — experiencing first-hand how unorganised data leads to inconsistent and unreliable counts.
What did I learn?	Ungrouped (raw) data is a list of individual data entries that has not yet been sorted or counted. A tally sheet organises data collection by using tally marks against pre-defined categories, making counting faster, more accurate, and reproducible. Designing a good tally sheet requires mutually exclusive categories and a clear research question.
How does this explain the phenomenon?	Groups designed and used their own tally sheets to collect at least 30 real data entries from classmates on a school-relevant question (favourite meal, transport mode, or sport). They connected this process directly to Mama Wanjiku's problem: she never had a tally sheet — she relied on memory — which is why her guesses about meal popularity were so wrong.

LESSON 3 (40 minutes): Finding the Middle Ground — Mean, Mode and Median for Mama Wanjiku	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners calculate the mean, mode and median from Mama Wanjiku's canteen frequency distribution table and use these three measures to give her a precise, data-driven recommendation about how many portions of each meal to prepare daily.
Knowledge	– Define mean as the sum of all values divided by the total number of values for ungrouped data. – Define mode as the value (or category) that occurs most frequently in a dataset. – Define median as the middle value when data are arranged in ascending order, and apply the formula (n+1)/2 to locate it. – Explain why different measures of central tendency can give different — yet equally valid — summaries of the same dataset.
Skills	– Calculate mean, mode and median from a frequency distribution table constructed in Lesson 2. – Select and justify the most appropriate measure of central tendency for a given real-life decision context.
Attitudes	– Demonstrate integrity by recording and using actual data values rather than guessing or rounding carelessly. – Appreciate that mathematical precision — not memory or intuition alone — leads to fairer, more effective decisions for community members like Mama Wanjiku.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	In Lessons 1 and 2 learners discovered that raw data is invisible information and built a frequency distribution table to reveal meal-choice patterns. Lesson 3 advances the storyline by extracting three numerical summaries — mean, mode and median — from that table, giving Mama Wanjiku the first concrete, mathematically grounded answer to her daily portion-planning problem.
Safety Notes	No specific hazards. Ensure learners handle rulers and pencils responsibly during table work.

B. LESSON OVERVIEW

This lesson sits at the heart of the evidence-gathering sequence. Having organised Mama Wanjiku's one-week canteen data into a frequency distribution table in Lesson 2, learners now squeeze three powerful numerical summaries out of that same table: the mean (average daily portions sold), the mode (the most frequently ordered meal), and the median (the middle value in an ordered dataset). Each measure answers a slightly different version of Mama Wanjiku's question — "How much should I cook?" — and learners must justify which measure is most useful for daily stock planning.

The lesson is structured around a concrete dataset: 120 learners recorded their daily meal choices across five days (Monday–Friday), producing daily ugali-with-sukuma-wiki totals of 54, 60, 57, 60 and 59. Learners calculate the mean (58 portions), identify the mode (60 portions — the most common daily total) and locate the median (58 portions). They then compare these values and discuss what each one tells Mama Wanjiku. A second, smaller dataset for githeri (the least popular meal: 8, 10, 7, 10, 10) is used for guided practice, reinforcing the procedures and deepening understanding of why the mode is not always the same as the mean.

Throughout the lesson, the anchoring phenomenon remains central: every calculation is framed as a direct answer to Mama Wanjiku's planning problem. The lesson closes with learners updating the class Driving Question Board (DQB) and revising their running causal model of "how data becomes a decision" to include the new layer of numerical analysis. Core competencies developed include critical thinking and problem solving (selecting the right measure) and communication and collaboration (justifying recommendations to peers).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the completed frequency distribution table from Lesson 2 — daily ugali-with-sukuma-wiki totals written prominently on the board: 54, 60, 57, 60, 59. Before any instruction, each learner silently writes in their exercise book: (1) their best guess at a single number that represents a 'typical' day's ugali sales, and (2) one sentence explaining their reasoning. Partners then share predictions for 90 seconds before three pairs report to the class.	Display the five daily totals (54, 60, 57, 60, 59) in large writing. Say: 'Mama Wanjiku needs to decide how many portions of ugali to cook tomorrow. Look at this week's data. In your exercise books, write ONE number you would advise her to cook — your best prediction — and explain in one sentence WHY you chose that number. You have 2 minutes. Work silently.' [Allow full 2 minutes.] Then say: 'Turn to your desk partner and compare your numbers. Do you agree? If not, who has the stronger reason?' [Allow 90 seconds.] Cold-call 3 pairs: 'Pair at the front left — what number did you choose and why?', 'Pair by the window — different answer?', 'Last pair — did you change your mind after talking?' Do NOT confirm or correct predictions yet. Record the range of predictions on the board (e.g., 57, 58, 59, 60) to return to after calculation. Wait 10–12 seconds after each pair's response before moving on.	Prediction-before-instruction (Elicit Prior Knowledge): By committing to a number before learning formal procedures, learners activate intuitive notions of 'average' or 'typical'. This surfaces misconceptions (e.g., always choosing the highest value, or just picking any number in the list) that the Explain Phase will directly address, making the formal definitions feel purposeful rather than arbitrary.	Teacher scans written predictions as learners share with partners. Key indicators: (1) Does the learner choose a number within the data range (54–60), suggesting reasonable number sense? (2) Does any learner spontaneously use language like 'add them up and divide' (nascent mean concept) or 'the one that appears most' (nascent mode concept)? Record 2–3 representative prediction strategies on the board. If all learners simply pick 60 (the highest), this signals a misconception to address explicitly in the Explain Phase.
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of	Working in pairs, learners receive a structured data card (or copy from the board) showing the full week's canteen dataset in table form: five days of ugali totals (54, 60, 57, 60, 59) and five days of githeri totals (8, 10, 7, 10, 10). They follow a step-by-step guided worksheet that leads them to (a) list the ugali values in ascending order, (b) add all five ugali values together and record the sum, and (c) count how many values there	Distribute the data cards. Say: 'Before we calculate anything, we are going to do what good mathematicians always do — organise our evidence carefully. Your worksheet has three steps. Step 1: Write the five ugali numbers in order from smallest to largest. Step 2: Add all five together and write the total. Step 3: Count how many numbers you have. That is all — do not divide yet.' [Allow 4 minutes.] Circulate	Structured Data Organisation (Ascending Order Protocol): Requiring learners to physically reorder the data before any calculation mirrors the scientific practice of organising evidence before drawing conclusions. Ascending order is not arbitrary — it is the prerequisite for finding the median and makes the mode visually obvious as a repeated value. This step connects directly back to Lesson 2's tally-sheet	Teacher checks two things during circulation: (1) Is the ascending order correct — 54, 57, 59, 60, 60? A common error is writing 54, 57, 60, 59, 60 (not fully sorted). (2) Is the sum 290? If multiple pairs get 291 or 289, pause the class for a shared re-addition before proceeding. These errors predict difficulties in the Explain Phase and must be resolved now.

Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	are. They do NOT yet calculate — they are gathering and organising evidence, paralleling Mama Wanjiku's own data-organisation journey.	and observe: check that learners write ascending order correctly (54, 57, 59, 60, 60) and that their sum is 290. If a learner writes 54+57+59+60+60 = 291, stop and ask: 'Check your addition — what is 54 plus 57?' rather than correcting directly. After 4 minutes, cold-call 2 learners to read their ordered list aloud. Say: 'Notice anything interesting about the ordered list? [Wait 12 seconds.] The number 60 appears TWICE. Keep that observation — it will matter in a moment.'	work: both are acts of imposing order on raw information.	
Explain Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Using their organised evidence from the Observe Phase, learners work through three calculations in sequence, guided by the teacher. (1) MEAN: They divide their sum (290) by the count (5) to get 58. (2) MODE: They identify 60 as the most frequent value (appears twice). (3) MEDIAN: They use the ordered list (54, 57, 59, 60, 60) and the locator formula position = (n+1)/2 = (5+1)/2 = 3rd value = 59 to find the median. After confirming all three values for ugali, learners independently apply all three procedures to the githeri dataset (8, 10, 7, 10, 10) as guided practice, then discuss with a partner: 'Which single number should Mama Wanjiku trust most for daily planning — mean, mode or median? Why?'	Write on the board: MEAN = sum ÷ count. Say: 'Take your sum — 290 — and divide by the number of values — 5. Calculate now.' [10-second wait.] Cold-call one learner: 'What did you get?' [Answer: 58.] 'So the MEAN daily ugali sales is 58 portions. Write a definition in your own words.' [1 minute.] Next, point to the ascending list: 54, 57, 59, 60, 60. Say: 'Which number appears more than once?' [Wait 10 seconds — do NOT answer for them.] 'That is the MODE — the most common value. Write: MODE = 60.' For the median, say: 'The MEDIAN is the middle value of an ordered list. I use the formula: position = (n+1) divided by 2. We have n = 5 values. So the median is at position... calculate it.' [Wait 12 seconds.] Cold-call: '(5+1)/2 = ?' [Answer: 3.] 'So the median is the 3rd value in our ordered list. Count along: 54 is 1st, 57 is 2nd, 59 is 3rd. MEDIAN = 59.'	Graduated Release with Parallel Dataset (I Do–We Do–You Do): The teacher models mean calculation with a think-aloud, guides mode and median as a whole class, then releases learners to apply all three procedures independently to the githeri dataset. Using a second real dataset from the same canteen table (rather than an abstract textbook example) keeps every calculation anchored to the phenomenon and prevents learners from feeling that statistics is a disconnected drill exercise.	Three checkpoints: (1) After mean: scan books — is 290 ÷ 5 = 58 written correctly? Flag learners who write 290 ÷ 5 = 57 or 59 (arithmetic slip). (2) After median: did learners correctly use position = (n+1)/2, or did they simply pick the middle number by eye? Both may produce the correct answer here but the formula approach is needed when n is even. Ask one learner to explain the formula aloud. (3) After githeri independent practice: circulate and check for three specific correct values: mean = 9, mode = 10, median = 10. If a learner gets median = 8 (second value instead of third), ask: 'How did you find the position of the middle value? Show me your formula.'

		Now pose the discussion question: 'Mama Wanjiku asks you: which number do I use — 58, 59 or 60? Turn to your partner. You have 90 seconds.' [Circulate and listen.] Cold-call 3 pairs. Then say: 'There is no single wrong answer here — but notice that the MODE (60) is the number she ran out of portions on — it is the most common HIGH day. If she cooks 60 every day she will rarely run short. The MEAN (58) is the mathematical average. A good statistician considers the purpose of the decision.' Leave both views on the board. For githeri independent practice: 'Now apply all three steps — order, sum, divide; spot the repeat; find the 3rd value — to the githeri data: 8, 10, 7, 10, 10. You have 5 minutes. Work alone first, then check with your partner.' [Circulate. Target answers: ascending = 7, 8, 10, 10, 10; sum = 45; mean = 9; mode = 10; median position = 3rd = 10.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Mean and Standard Deviation of Discrete Random Variables]	Each learner writes one sticky note (or a slip of paper) responding to ONE of three colour-coded prompts: GREEN — 'Something I now know that helps answer the driving question'; ORANGE — 'A new question this lesson raised for me'; RED — 'Something I am still confused about.' Learners post their notes on the class DQB under the heading 'Lesson 3: Mean, Mode, Median.' The teacher then reads 2–3 notes aloud and connects them explicitly back to the driving question: 'How can we turn raw data into statistics that help Mama Wanjiku solve real	Say: 'Our driving question is: How can we turn raw data from our school and community into statistics that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems? Think about what we did today. Take a sticky note — green, orange, or red — and write your response to the matching prompt on the board. You have 2 minutes. Write one clear sentence.' [Distribute notes / slips. Allow 2 minutes of silent writing.] As learners post notes, read 2–3	Driving Question Board Update (Public Knowledge Construction): The DQB serves as a cumulative, visible record of the class's evolving understanding. By colour-coding responses (know / wonder / confused), the teacher gains real-time formative data across all three categories and communicates to learners that new questions are as valuable as new answers. Connecting posted notes back to the driving question reinforces the purpose of every calculation.	Teacher reads all posted notes during or after the lesson. Key indicators: (1) GREEN notes should reference at least one of the three measures by name and connect it to a decision (e.g., 'mode of 60 means Mama Wanjiku should cook at least 60 ugali portions'). Vague notes ('I learned about averages') signal shallow understanding. (2) ORANGE/RED notes reveal which concepts need reinforcement — if more than 4 learners write confusion about when to use median vs mean, open that comparison explicitly at the start of Lesson 4.

[(Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	problems?'	aloud: 'This learner says: the mean of 58 portions helps Mama Wanjiku order the right amount of maize flour. That directly answers our driving question — she now has a NUMBER, not a guess.' Read an orange note: 'This learner asks: what if sales change in a different term? Great question — we will return to that in later lessons.' Read a red note without identifying the writer: 'This learner is still unsure when to use median vs mean. We will build on that tomorrow.' Point to the DQB: 'Notice how our board is growing — each lesson adds new answers AND new questions. That is exactly how real statisticians work.'		
Model Building Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their individual causal-model diagrams started in Lesson 1 (a flow diagram showing how data becomes a decision). They add a new layer: a box labelled 'Calculate Measures of Central Tendency' between 'Frequency Distribution Table' and 'Make a Decision / Graph.' Inside this new box they write three branches: MEAN (sum ÷ count), MODE (most frequent), MEDIAN (middle value). They annotate each branch with the ugali result (58, 60, 59) and a one-word note on what it tells Mama Wanjiku ('average', 'peak day', 'typical day'). Two learners share their updated models with the class for 1 minute each.	Say: 'In Lesson 1 your model showed: Raw Data → Frequency Table → Pattern Visible. In Lesson 2 you added: Tally Sheet → Organised Counts. Today we add a new layer. Open your model diagrams. Draw a new box after Frequency Table and label it: Calculate Measures of Central Tendency. Add three branches inside: MEAN, MODE, MEDIAN. For each branch, write the ugali number we calculated today and one word describing what it tells a decision-maker.' [Allow 4 minutes.] After 3 minutes, say: 'I need two volunteers to hold up their model and explain their new layer in 30 seconds each.' Select one stronger and one developing learner deliberately. After each shares, ask the class: 'Does anyone have a different label for what mean / mode / median tells us? [Wait 10 seconds.] Good — there may be more than one valid	Cumulative Causal Model Revision (Running Explanation): The flow diagram is the learners' personal, growing theory of how statistical tools transform raw data into decisions. Each lesson adds a new mechanism to the diagram. This practice mirrors how scientists build explanatory models incrementally — each new piece of evidence adds a layer rather than replacing the whole. Requiring specific numerical annotations (58, 60, 59) prevents the model from remaining abstract.	Circulate during model-building and look for three things: (1) Is the new box correctly positioned between 'Frequency Table' and 'Decision / Graph' — indicating the learner understands the logical sequence of statistical analysis? (2) Are all three measures (mean, mode, median) present with correct numerical values? (3) Are the one-word annotations meaningful (e.g., 'average', 'peak', 'middle') rather than just restating the numbers? Learners whose models show correct sequence and all three values with annotations are meeting expectations. Those who add the box but leave branches empty or incorrect need targeted follow-up.

		description.' Close by connecting forward: 'Next lesson we will take these numbers and turn them into pictures — bar graphs and pie charts — so that even someone who does not read tables can instantly see what the data says. Our model will grow again.'		

D. TEACHER REFLECTION
1. Did all learners successfully apply the (n+1)/2 median-position formula, or did most rely on counting by eye? If the formula was not secure, how will I build it into the opening of Lesson 4 before moving to bar graphs?\n2. During the Explain Phase discussion ('Which measure should Mama Wanjiku trust most?'), how many learners could articulate a reason beyond 'it is bigger' or 'it is the middle'? What question stems could I use next time to push for more precise justification?\n3. Were the learners who struggled with addition in Lesson 2 (summing tally marks) also the ones who made arithmetic errors when computing the mean today? What targeted numeracy support can I put in place before Lesson 4?\n4. Looking at the DQB orange and red notes: which confusion or new question appeared most frequently? Does it need addressing at the start of Lesson 4, or can it be resolved through the graphing activity?\n5. Were the causal model diagrams genuinely cumulative — did learners connect today's layer to the previous two — or did they treat the model as a separate worksheet? How can I make the continuity between lessons more explicit in my instructions next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that the five daily ugali sales totals (54, 60, 57, 60, 59) are not all the same — they vary from day to day — and that the number 60 appears twice, making it stand out as the most common (peak) day. They also observed that ordering the data (54, 57, 59, 60, 60) makes both the middle value and the repeated value immediately visible.
What did I learn?	Three measures of central tendency summarise ungrouped data in different but complementary ways: the MEAN (290 ÷ 5 = 58) gives the mathematical average; the MODE (60) identifies the most frequently occurring value; and the MEDIAN — found at position (n+1)/2 = 3rd value = 59 — gives the middle value of the ordered dataset. Each measure answers Mama Wanjiku's planning question slightly differently: mean gives a balanced daily target, mode reveals the peak demand day, and median offers a robust middle estimate.
How does this explain the phenomenon?	Mama Wanjiku's memory-based guesses failed because human memory tends to anchor on recent or dramatic events rather than calculating a true average across all days. The mean of 58 portions per day is a precise, reproducible number — derived from the full week's evidence — that removes the distortion of memory. The mode of 60 further reveals that on the most common type of day she needs even more portions than the mean suggests. Together, these statistics transform the noisy, unorganised raw data from Lesson 2 into a concrete, mathematically justified daily cooking target — exactly the kind of decision-support tool that Mama Wanjiku, and any community decision-maker, needs.

LESSON 4 (40 minutes): From Chaos to Clarity: Building Frequency Distribution Tables	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct frequency distribution tables for ungrouped data and discover how organising raw data reveals patterns invisible to memory — directly solving Mama Wanjiku's canteen puzzle.
Knowledge	A frequency distribution table organises raw ungrouped data by listing each data value and the number of times it occurs (its frequency), making patterns visible and enabling further statistical computation.
Skills	Draw a frequency distribution table for ungrouped data; use tally marks to count occurrences accurately; identify the highest and lowest frequencies; read and interpret a completed frequency distribution table.
Attitudes	Appreciate that systematic organisation of data — rather than memory or guesswork — is the honest and reliable foundation for decision-making; display integrity when recording tallies.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	In Lesson 1 learners collected raw meal-order data. In Lessons 2–3 they explored the anchoring phenomenon and the supporting transport phenomenon. Now, in Lesson 4, they apply the first statistical tool — the frequency distribution table — to their own raw data sets, mirroring exactly what a real analyst would do for Mama Wanjiku. The table transforms their noisy lists into readable evidence and sets up Lessons 5–6 (mean, mode, median) by raising the question: 'Can one number represent the whole table?'
Safety Notes	No physical hazards. Remind learners to handle shared data sheets carefully and return them intact. Ensure respectful group discussion — no mocking of another group's data or tally errors.

B. LESSON OVERVIEW

Learners retrieve the raw meal-order data they collected in Lesson 1. The teacher first models frequency distribution table construction using the supporting transport phenomenon (school-gate tally data from Mwangaza Secondary School). Groups then construct their own frequency distribution tables for their canteen data sets. A structured gallery comparison reveals how organisation exposes patterns — especially ugali's dominance — that were invisible in the raw list. The lesson closes with a DQB update that bridges to measures of central tendency in Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES	Each learner receives their group's raw data sheet from Lesson 1 — a disorganised list of 120 meal choices written in the order they were collected. Individually and silently, learners study the list for 90 seconds and write two	Distribute raw data sheets face-down. Say: 'Do not flip the sheet yet. In a moment I will ask you to look at this list of 120 meal choices — the same kind of list Mama Wanjiku would have if she ever wrote down her orders. I want you	Prediction Activation — surfacing prior intuition before instruction so learners have a personal stake in the outcome. By committing a prediction to paper, each learner creates cognitive dissonance they will resolve through the	Observable indicator: Every learner has written a meal name AND a confidence rating in their exercise book before sharing. Teacher scans books during circulation. If more than one-third write 'just guessing', note this

for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	predictions in their exercise books: (a) Which meal appears most often? (b) How confident are you in your answer — very confident, somewhat confident, or just guessing? Learners share predictions with a shoulder partner before whole-class cold-call.	to predict: which meal comes up most often, and how confident are you?' Flip instruction: 'Flip your sheet. You have 90 seconds — go.' [Start timer. Circulate silently. Do NOT answer questions.] After 90 seconds: 'Pens down. Write your prediction and confidence level.' Cold-call exactly 5 learners using name cards: ask each, 'What did you predict, and why?' [WAIT TIME: 12 seconds after each name before prompting.] Tally predictions on the board in a simple two-column list: Meal | Number of classmates who predicted it. Do not confirm or deny any prediction — say: 'We will let the data speak for itself.'	observation phase, deepening engagement with the table-building process.	aloud: 'Many of us are guessing — that is exactly Mama Wanjiku's problem. Let us see if a table can do better than a guess.'
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher projects (or draws on the board) the school-gate transport data from Mwangaza Secondary School: a raw list of 40 entries — matatu, boda boda, walking, matatu, cycling, walking, matatu… — collected at the school gate one Monday morning. Learners watch and participate as the teacher live-models constructing a frequency distribution table step by step using this transport data. Learners copy the completed model table into their books. They then open their group's canteen raw data sheet and — using the model as a guide — begin constructing their own frequency distribution table in groups of four.	Say: 'Before we tackle our canteen data, I am going to show you the tool we need — using a different data set so you can see the steps clearly.' Draw the raw transport list on the board. Step 1 — Column headers: 'I write three columns: Mode of Transport | Tally | Frequency. Copy this.' Step 2 — First pass through the data: 'I read each entry aloud. Every time I read it, I make one tally mark next to that category. Watch my hand.' Read all 40 entries aloud at a steady pace, marking tallies visibly. Step 3 — Grouping tallies in fives: 'After every four marks I cross through to make a bundle of five — this makes counting easier.' Step 4 — Counting frequencies: 'I count each bundle and write the number in the Frequency column.' Step 5 — Check: 'I add all frequencies: they must sum to 40 — our total number of entries. If they do not, I have made an error.' [Pause.] Ask the class: 'Why must the frequencies add up to 40?'	Live Modelling with Structured Role Assignment — the teacher makes the invisible thinking process of table construction visible through narrated step-by-step action. Assigning group roles (reader, tally marker, frequency counter, checker) distributes cognitive load and ensures every learner is actively processing data rather than watching one peer do the work. The transport supporting phenomenon serves as a low-stakes rehearsal before learners apply the tool to their own canteen data.	Observable indicators: (1) Each group's tally column shows clear bundles of five with a crossing stroke. (2) The sum of frequencies in every completed table equals exactly 120. (3) When asked Q1–Q3 during circulation, at least one group member can respond without prompting. Teacher notes on a clipboard which groups cannot explain the connection between their highest frequency and Mama Wanjiku's problem — these groups receive a follow-up question in the Explain phase.

		[WAIT TIME: 10 seconds.] Cold-call 2 learners. Then say: 'Now, using this model, your group will build a frequency distribution table for your canteen data. You have 12 minutes. Roles: one person reads entries aloud, one marks tallies, one counts and writes frequencies, one checks the total equals 120.' Circulate. Stop at each group and ask one of three targeted questions — rotate so every group hears at least one: Q1: 'What is the highest frequency in your table so far?' Q2: 'What does that tell us about Mama Wanjiku's problem?' Q3: 'Could Mama Wanjiku have seen this pattern without the table — why or why not?' [WAIT TIME: 12 seconds after each question before accepting an answer.] If a group's frequencies do not sum to 120, say: 'Your total is [X]. What should it be? Where might the error be hiding?' — do not correct it for them.		
Explain Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups post their completed frequency distribution tables on the wall (or hold them up). The class conducts a silent gallery walk for 3 minutes, reading every group's table and writing in their books: (a) the most frequent meal in each table, (b) one similarity and one difference between tables. Whole-class discussion follows, connecting findings to Mama Wanjiku's situation. Learners then individually answer two written consolidation questions.	Say: 'Post your table on the wall. We are going to do a silent gallery walk — no talking, just reading and writing. You have 3 minutes.' [Set timer. Circulate and observe what learners write.] After the gallery walk, bring the class together. Ask: 'Raise your hand if ugali with sukuma wiki had the highest frequency in the table you read.' [Count hands aloud.] 'Raise your hand if githeri had the lowest frequency in at least one table.' [Count hands.] Then cold-call 3 learners — one who raised their hand for ugali, one for githeri, one who noticed a difference between tables: 'Tell us what you noticed and what it means for Mama Wanjiku.' [WAIT TIME: 12 seconds	Gallery Walk with Evidence-Based Discussion — comparing multiple groups' tables makes the consistency of the pattern (ugali's dominance) visible across different data subsets, building the idea that a well-constructed table reveals objective truth rather than individual memory. The consolidation questions require learners to move from description ('ugali is most frequent') to application ('here is what Mama Wanjiku should do'), operationalising NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data) and Practice 6 (Constructing Explanations).	Observable indicators: (1) In CQ1, learner explicitly cites a frequency number from their table, not just a meal name. (2) In CQ2, learner connects the table to a concrete decision (e.g., 'order 58 portions of ugali ingredients per day'). Teacher collects exercise books of 6 learners (2 per ability level, selected in advance) at end of lesson to check written responses against these indicators.

		each.] After responses, anchor the explanation: 'Every group's table is telling the same story — even though they are made from different subsets of the 120 orders. Ugali with sukuma wiki appears roughly half the time. Mama Wanjiku's memory told her something different on some days. Why?' [WAIT TIME: 10 seconds.] Cold-call 2 learners. Accept and build on responses; guide toward: 'Memory is selective and can be wrong — a frequency table cannot lie about what was recorded.' Then write on the board: 'A frequency distribution table organises raw data so that patterns become visible.' Say: 'Write this in your books as today's key statement.' Now pose the two consolidation questions — learners answer individually in silence for 4 minutes: CQ1: 'Using your group's frequency distribution table, state the meal Mama Wanjiku should prepare the most of each day. Explain using your frequency figures.' CQ2: 'If Mama Wanjiku had kept a tally every day for one week, how would her frequency table help her order ingredients on Monday morning for the week ahead?' Cold-call 4 learners to share answers — 2 per question. [WAIT TIME: 12 seconds each.] Affirm correct reasoning; redirect incomplete answers with: 'You said [X] — what number in your table supports that?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ungrouped Data to Find the Mean](#)	Each group receives one sticky note. They write one new question their frequency distribution table raised — something they still cannot answer — and post it on the class DQB. The teacher reads all new questions aloud and	Say: 'Our table tells us which meal is most popular and which is least popular. But Mama Wanjiku also wants to know: what is a typical day? If she had to pick one number to represent the whole week's ugali orders, what would it	Question-Driven DQB Update — requiring each group to formulate a question (rather than a statement) ensures learners are operating at the edge of their current understanding. Clustering questions by type makes the	Observable indicator: Each group's sticky-note question is genuinely about a gap in their understanding (e.g., 'How do we find the middle value of our data?' or 'What if two meals have the same frequency — which one is typical?') rather than

[Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	frames the bridge to Lesson 5.	be? Talk to your group for 60 seconds — write your best question about this on your sticky note and post it on the DQB.' [Circulate during 60-second discussion. Listen for language like 'average', 'middle number', 'most common number' — these show readiness for Lesson 5.] After posting: Read each sticky note aloud. Organise them into clusters on the board: 'Questions about a typical value' vs 'Questions about why patterns differ'. Point to the cluster about typical values: 'These questions are asking about what mathematicians call measures of central tendency — the mean, the mode, and the median. That is exactly where we are going in our next lesson.' Write on the DQB banner strip: 'A frequency table organises data — but which single number best represents the whole data set?' Say: 'This is our new driving question. Keep it in mind tonight.' Do NOT answer the question — preserve productive uncertainty.	logical progression of the unit visible to learners and positions Lesson 5 as a natural, motivated next step rather than an arbitrary new topic.	a procedural question about the table. Teacher sorts sticky notes after class: questions that are still about table construction indicate those groups need consolidation at the start of Lesson 5.
Model Building Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar	Each group retrieves their running statistical model — a large sheet of paper started in Lesson 2 showing Mama Wanjiku's problem and the data they have gathered. They add a new layer: sketch their completed frequency distribution table (or paste a miniature copy) onto the model sheet, draw an arrow from the 'raw data' box to the 'frequency table' box, and annotate the arrow with the phrase: 'Organisation reveals pattern.' Groups then add one speech bubble for Mama Wanjiku on their model: what she now knows that she did not know before.	Say: 'In Lesson 2 you started a model of Mama Wanjiku's problem. Today you have a new tool to add. Take 5 minutes to update your model: add your frequency table, draw the arrow from raw data to the table, and write the annotation — Organisation reveals pattern. Then add Mama Wanjiku's speech bubble: what does she now know?' [Distribute model sheets. Circulate. Prompt groups who are only copying the table: 'What does the arrow mean? Why does it go that direction?' WAIT TIME: 10 seconds.] After 4 minutes, cold-call 2 groups to share their Mama	Cumulative Model Revision — adding to a persistent visual model (rather than starting fresh each lesson) makes the progression of statistical reasoning concrete and cumulative. The annotation requirement ('Organisation reveals pattern') forces learners to articulate the epistemological insight of the lesson — that the tool, not just the data, produces understanding. The Mama Wanjiku speech bubble maintains emotional connection to the phenomenon and ensures the model remains anchored to a real-world decision-making context.	Observable indicators: (1) The arrow on the model runs from 'raw data list' to 'frequency table' — not the reverse. (2) The speech bubble contains at least one specific number from the frequency table. (3) The annotation reads 'Organisation reveals pattern' or an equivalent learner-generated phrase. Teacher photographs each group's updated model at end of lesson for portfolio evidence and to track model sophistication across the 8-lesson sequence.

readings		Wanjiku speech bubble aloud. Affirm responses that include a specific frequency ('She now knows that about 58 learners choose ugali every day'). Close by saying: 'Your model is growing. Next lesson we add the next layer — a single number that summarises the whole table. See you then.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) Prediction accuracy — did learners' pre-table predictions differ significantly from their table results? If yes, did you make this discrepancy explicit and connect it to Mama Wanjiku's memory-based guessing? (2) Tally accuracy — how many groups produced a frequency sum that did not equal 120 on first attempt? What error patterns emerged (missed entries, double-counting)? Plan a 3-minute error-analysis routine at the start of Lesson 5 if more than two groups had summing errors. (3) DQB questions — did the sticky-note questions cluster around 'typical value' / central tendency? If most groups asked procedural questions about the table rather than conceptual questions about representation, spend the first 5 minutes of Lesson 5 revisiting the key statement: 'A table tells us frequencies — but which single number speaks for the whole data set?' (4) Equity check — during the gallery walk and cold-calls, which learners did not speak? Ensure those learners are targeted for cold-call in Lesson 5. (5) Model building — were Mama Wanjiku's speech bubbles specific (citing numbers) or vague ('she knows more now')? Vague responses suggest learners have not yet connected the table to decision-making; address this at the start of Lesson 5 before introducing mean, mode and median.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that when we sorted our raw list of 120 canteen meal orders into a frequency distribution table — listing each meal and tallying how many times it appeared — the data stopped looking like noise and started showing a clear pattern. In every group's table, ugali with sukuma wiki had the highest frequency (around 55–60 out of 120), while githeri had the lowest. The same pattern appeared in the transport data the teacher modelled: once tallied, the most and least common modes of transport became immediately obvious.
What did I learn?	We learned that a frequency distribution table organises ungrouped raw data by recording each possible value and its frequency (how often it occurs), using tally marks grouped in fives for easy counting. The sum of all frequencies must equal the total number of data entries — this is our accuracy check. Organisation reveals patterns that are invisible in a raw list, and it was this kind of organisation that Mama Wanjiku was missing when she relied on memory alone.
How does this explain the phenomenon?	We can now explain why Mama Wanjiku's memory-based guesses failed: she was trying to process 120 individual orders mentally, without any system for counting or recording. A frequency distribution table does what memory cannot — it counts every single entry without bias, groups them by category, and displays the pattern clearly. With a completed table, Mama Wanjiku can see at a glance that she needs roughly 58 portions of ugali with sukuma wiki every day and far fewer portions of githeri, allowing her to order ingredients accurately, reduce waste, and ensure every learner gets served. The table does not yet tell us what a 'typical' day looks like as a single number — that is the question we carry into Lesson 5.

LESSON 5 (40 minutes): Picture This — Representing Mama Wanjiku's Data as Bar Graphs, Line Graphs and Pie Charts	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners represent Mama Wanjiku's canteen frequency data visually using bar graphs, line graphs and pie charts, discovering how each graph type reveals a different pattern in the same dataset.
Knowledge	– A bar graph uses rectangular bars whose heights represent frequencies; bars are separated and can be drawn vertically or horizontally. – A pie chart uses sectors of a circle where the angle of each sector = (frequency ÷ total) × 360°; it shows proportion rather than count. – A line graph connects data points with straight lines and is most useful for showing how a quantity changes over time or sequence. – The same dataset can be represented by different graph types, and the choice of graph depends on what question you want to answer.
Skills	– Draw accurately scaled bar graphs, line graphs and pie charts from a given frequency distribution table. – Calculate sector angles for a pie chart and use a protractor and ruler to construct it to scale. – Interpret and compare information presented in different graph formats to draw conclusions.
Attitudes	– Appreciate that visual representation transforms numbers into stories that decision-makers — like Mama Wanjiku — can act on immediately. – Demonstrate honesty and integrity by drawing graphs accurately from real data rather than distorting scales to mislead.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Having built a frequency distribution table (Lesson 4) and computed mean, mode and median (Lesson 3), learners now give Mama Wanjiku a visual tool — graphs and charts — that instantly communicates the meal-choice patterns she could never see by memory alone, advancing the class toward a complete statistical report for the canteen.
Safety Notes	No specific hazards. Ensure learners handle rulers, protractors and pencil compasses carefully to avoid injury. Remind learners not to press compass points onto desks.

B. LESSON OVERVIEW

This lesson is the visual payoff of the entire evidence-gathering sequence. Learners have already collected raw data, organised it into a frequency distribution table, and calculated measures of central tendency for Mama Wanjiku's canteen data. Now they face a new challenge: Mama Wanjiku cannot read a table of numbers at a glance during the busy lunch rush — she needs a picture. In this lesson learners transform the same frequency table (ugali with sukuma wiki: 56, chapati: 30, githeri: 14, rice and beans: 28, mandazi: 22; total = 150 orders recorded over one week) into three different visual representations — a bar graph, a pie chart, and a line graph showing daily ugali sales across five school days — and then compare what each graph reveals.

The lesson opens with a prediction challenge: learners are shown a blank set of axes and asked which graph type would best help Mama Wanjiku decide how much ugali to cook tomorrow. This surfaces prior knowledge about graphs and creates productive disagreement. The observation phase involves a structured, step-by-step construction activity where learner pairs draw all three graph types using the canteen dataset, following a scaffolded worksheet. The explain phase uses a gallery-walk and whole-class discussion to compare the three graphs side by side, naming what each one shows that the others do not.

Throughout the lesson, teacher moves are tightly sequenced: precise wait times after questions, cold-call turns, and targeted questioning force learners to connect every graph back to the anchoring phenomenon. By the end of the lesson learners can articulate not just how to draw graphs but why visual representation is a decision-making tool — and they can point to exactly which graph would have prevented Mama Wanjiku's ugali shortage and githeri waste.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mixed Numbers in Applications: Sharing a Chocolate Bar (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown three blank graph frames on the board — one with separated bars, one with a circle, one with connected points — without labels or data. Working individually in their exercise books for 2 minutes, they write: (1) the name of each graph type, (2) one thing each graph type is good at showing, and (3) which graph they would hand to Mama Wanjiku first and why. They then share predictions with a seat-partner before a whole-class cold-call round.	Display the three blank frames and say: 'Before we draw anything, I want you to think like a graphic designer hired by Mama Wanjiku. She has 30 seconds at the start of each day to look at one chart. Which of these three frames would give her the most useful picture of her canteen data — and why? Write your answer in your book. You have 2 minutes — go.' [Allow 2 minutes of silent individual writing. Circulate, read over shoulders, note interesting predictions without commenting yet.] After 2 minutes: 'Turn to your partner and compare your choice. Do you agree? If not, try to convince each other. 1 minute — go.' [Allow 1 minute.] Cold-call 4 pairs in sequence — choose one pair who agreed and one who disagreed: 'Amina and Brian, you agreed — which graph did you choose and what was your reason?' [WAIT 10 seconds after each response before moving on.] After 4 responses, say: 'Hold those predictions. By the end of this lesson you will each be able to defend your choice with evidence — not just a feeling.'	Anticipation Prediction with Justification — learners commit to a reasoned choice before instruction, activating prior graph knowledge from lower school and creating cognitive dissonance that motivates the lesson. Requiring a written justification (not just a circle-the-answer) forces elaboration and makes thinking visible for the teacher's formative read.	Teacher circulates during the 2-minute individual writing phase and scans books for: (a) correct naming of all three graph types — if more than a third of the class cannot name 'pie chart' or 'line graph', pause and do a quick 30-second naming activity before proceeding; (b) quality of justification — vague answers ('bar graph looks nice') versus substantive ones ('bar graph shows which meal is biggest'). Teacher mentally notes which learners have strong prior knowledge to use as explainers during the Explain phase, and which learners are uncertain to cold-call purposefully during Observe.
Observe Phase 📹 **VIDEO:** Video: Energy	Working in pairs, learners receive a printed or teacher-drawn data card showing the canteen frequency table (ugali with sukuma	Distribute data cards. Say: 'You have the same data Mama Wanjiku's helpers collected — 150 meal orders over one week. Your	Scaffolded Construction with Checkpoint Pauses — rather than letting learners loose to draw all three graphs simultaneously, the	At each checkpoint the teacher does a 'show-of-books' scan — all learners hold up their book open to the current graph. Teacher checks:

[Skate Park: Basics](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Mixed Numbers in Applications: Sharing a Chocolate Bar (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	wiki: 56, chapati: 30, githeri: 14, rice and beans: 28, mandazi: 22; total = 150) and a second card showing daily ugali sales for Monday–Friday (54, 60, 55, 62, 59). Using their exercise books, a ruler, protractor and pencil compass, each pair constructs all three graphs step by step, guided by teacher-led 'checkpoint pauses' after each graph. For the pie chart, learners first calculate each sector angle using the formula (frequency ÷ 150) × 360° before drawing. For the line graph, learners plot the five daily ugali figures and connect points. Each graph is given a title, labelled axes, and a key where needed.	job is to draw three graphs from this data. We will go graph by graph together.' [CHECKPOINT 1 — Bar Graph] 'First, the bar graph. In your book, draw two axes. The horizontal axis will show meal names; the vertical axis will show frequency. Choose a scale — what is the highest frequency? [Cold-call 1 learner.] 56 — so what is a sensible scale going up to at least 60? Discuss with your partner for 30 seconds.' [WAIT 30 seconds.] 'Let us agree on a scale of 0 to 60, going up in 10s. Draw five separated bars. Label each one. Give your graph a title: Canteen Meal Orders — One Week. Go — you have 4 minutes.' [Circulate, check bar width consistency and separation, labelling.] After 4 minutes, do a show-of-books scan. [CHECKPOINT 2 — Pie Chart] 'Now the pie chart — this is where we need our maths. Before you draw anything, calculate the sector angle for each meal. The formula is: angle = frequency divided by total, then multiply by 360. Work out all five angles in your book. You have 3 minutes.' [WAIT, circulate, check arithmetic.] Cold-call 1 learner: 'Wanjiku, what angle did you get for ugali?' [WAIT 10 seconds.] 'Correct — 56 ÷ 150 × 360 = 134.4°, which we round to 134°. Compare that with your partner.' After all 5 angles are verified sum to 360°, say: 'Now draw your circle, use your protractor to mark each sector, shade and label. 5 minutes — go.' [CHECKPOINT 3 — Line Graph] 'Last graph — daily ugali sales across Monday to Friday. Plot the five points from your second data card, connect them with straight lines. What goes on the horizontal	teacher breaks construction into three sequential checkpoints with a class-wide pause and verification after each. This prevents compounding errors (a wrong scale on the bar graph would corrupt all subsequent work) and gives the teacher three precise moments to assess the whole class and redirect before moving on. The strategy also models the professional data-visualisation workflow: calculate first, then draw.	(1) Bar graph — are bars separated (not touching like a histogram)? Is the vertical axis labelled with units (number of orders)? Is there a title? (2) Pie chart — do the five sector angles sum to 360°? Has the learner shown the calculation working, not just the drawing? (3) Line graph — are points plotted at correct frequencies? Are points connected by straight lines (not curves)? Are axes labelled? Any learner whose pie-chart angles do not sum to 360° is redirected immediately: 'Check your arithmetic for githeri — what is 14 ÷ 150 × 360?'

		axis? [Cold-call 1 learner.] Days of the week. What goes on the vertical axis? [Cold-call 1 learner.] Number of portions. 3 minutes — go.'		
Explain Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mixed Numbers in Applications: Sharing a Chocolate Bar (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Pairs post their completed bar graph and pie chart on the classroom walls (line graphs remain in books). The class does a 4-minute silent Gallery Walk, moving around the room reading other pairs' graphs and writing one observation on a sticky note per graph they visit. Back in seats, learners answer three focus questions in their books: (1) Which graph most clearly shows that ugali is the dominant meal choice? (2) Which graph would best help Mama Wanjiku spot a trend — is ugali demand rising or falling across the week? (3) If Mama Wanjiku could only print one chart to stick above her cooking station, which would you recommend and why? A whole-class discussion follows, led by targeted cold-calls, before learners write a 3-sentence conclusion connecting graphs back to the phenomenon.	After the gallery walk, bring the class back to seats. Say: 'You have now looked at many versions of the same data. Let us think carefully about what each graph does.' Ask: 'Question 1 — which graph made it instantly obvious that ugali is the most popular meal — the bar graph or the pie chart? Discuss with your partner for 30 seconds.' [WAIT 30 seconds.] Cold-call 2 learners with opposing views: 'Otieno, you said bar graph — explain your reasoning.' [WAIT 10 seconds after response.] 'Fatuma, you said pie chart — do you agree with Otieno or do you have a different reason?' [WAIT 10 seconds.] Affirm both as valid, then push deeper: 'The bar graph shows the size difference clearly — the ugali bar is nearly 4 times the githeri bar. The pie chart shows that ugali is 37% of ALL orders — nearly 2 out of every 5 meals. Both facts are useful — but for different questions.' Then ask: 'Question 2 — can you use the bar graph or the pie chart to answer the question: is ugali demand going UP or going DOWN across the week?' [Cold-call 1 learner.] [WAIT 10 seconds.] Establish: 'No — because neither of those graphs shows TIME. That is why we drew the line graph. The line graph is the only one that shows the Monday-to-Friday trend.' Finally: 'Write three sentences in your book. Sentence 1: Which graph you recommend for Mama Wanjiku's cooking station and why.	Compare-Contrast Discussion with Anchored Conclusion Writing — the gallery walk builds a shared visual reference across the whole class, while the three focus questions create a compare-contrast cognitive structure that forces learners to articulate functional differences between graph types. The final 3-sentence written conclusion anchors the abstract graph-comparison discussion directly back to the canteen phenomenon, ensuring that graph literacy is always framed as a decision-making tool rather than a drawing exercise.	Teacher reads the 3-sentence written conclusions over learners' shoulders during writing time. Look for: (1) Sentence 1 — does the learner give a functional reason ('the pie chart shows proportions clearly') rather than aesthetic preference ('the pie chart looks nicer')? (2) Sentence 2 — does the learner make a specific causal claim about Mama Wanjiku's problem ('she would have known to cook at least 56 portions of ugali each day instead of guessing')? (3) Sentence 3 — does the learner correctly read the trend from the line graph (ugali demand fluctuates between 54 and 62 with a peak on Wednesday, showing moderate variation but no strong upward or downward trend)? Learners who write only one sentence or who copy from neighbours are flagged for follow-up questioning in the DQB phase.

		Sentence 2: What would have happened if Mama Wanjiku had seen this graph at the start of the week instead of relying on memory. Sentence 3: What the line graph tells us about whether demand for ugali is stable, growing or falling.' [Allow 3 minutes for writing.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mixed Numbers in Applications: Sharing a Chocolate Bar (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front. Each learner writes one new sticky note responding to the prompt: 'After today's graphs, what new question do you now have about how statistics and graphs help decision-makers in Kenya?' Learners place their sticky notes on the board under one of three column headers: WHAT WE NOW KNOW (green), WHAT WE ARE STILL WONDERING (yellow), WHAT WE WANT TO INVESTIGATE NEXT (pink). The class reads the board silently for 1 minute and the teacher facilitates a 2-minute whole-class discussion to identify one emerging question that the class agrees to investigate in Lesson 6.	Say: 'We have now built frequency tables, calculated mean, mode and median, and drawn three types of graph — all from Mama Wanjiku's canteen data. Look at our Driving Question at the top of the board: How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems? Take 1 minute to write your sticky note. Think: what can you now answer that you could not answer in Lesson 1? And what new question has today's work opened up for you?' [Allow 1 minute for writing.] 'Place your note in the correct column.' [Allow 30 seconds for placement.] After placement, scan the yellow column aloud: read 2–3 'still wondering' notes verbatim without editing. Say: 'I see several of you are asking: can these same graphs mislead people if drawn badly? And others are asking: how do we compare canteen data from two different schools? These are excellent statistical thinking questions. Let us hold them — they will guide Lesson 6 and Lesson 7.' Point to the pink column: 'Three of you want to draw graphs on a computer or phone — that is a real-world data literacy skill. We will visit that in Lesson 7.' Record the class's agreed	Three-Column DQB Sorting — the three-column structure (Know / Wondering / Investigate) transforms the DQB from a passive display into an active metacognitive tracking tool. By physically placing a sticky note, each learner makes a personal claim about the state of their own understanding. The teacher's real-time reading of the 'still wondering' column gives authentic formative data about whole-class gaps and surfaces genuine learner-generated questions that can anchor the next lesson — making the storyline feel driven by learner curiosity rather than a predetermined script.	Teacher reads sticky notes during placement and mentally tallies: how many learners are placing notes in 'KNOW' versus 'WONDERING'? A class where the majority of notes are in the green 'KNOW' column may indicate surface-level confidence without genuine understanding — probe 2–3 of those learners verbally: 'You say you know how to interpret a pie chart — tell me what Mama Wanjiku's pie chart says about githeri.' A class where most notes are in the yellow 'WONDERING' column signals that consolidation is incomplete and that the opening of Lesson 6 needs to revisit today's key concepts before introducing new content. Flag any learner who cannot produce a sticky note at all for one-on-one check-in after class.

		emerging question on the board in a permanent marker box: 'How do we interpret and compare graphs — and how can a badly drawn graph lie?'		
Model Building Phase 📹 **VIDEO:** [Video: Energy Skate Park: Basics] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mixed Numbers in Applications: Sharing a Chocolate Bar (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners turn to the cumulative 'Canteen Statistics Model' page in their exercise books — a running annotated diagram they have been building since Lesson 1. Today they add a new section titled 'VISUAL REPRESENTATIONS' and sketch miniature versions of all three graphs they drew today, labelling each with: the graph type, what it is best used for, and one key insight it gives about Mama Wanjiku's canteen. Below the sketches they write one updated answer to the driving question, explicitly stating how today's graphs advance the solution. Learners who finish early add a 'LIMITATIONS' box noting one weakness of each graph type.	Say: 'Turn to your Canteen Statistics Model page. Since Lesson 1 you have been building a picture of how statistics transforms noise into knowledge. Today we add the visual layer. Add a new section — title it VISUAL REPRESENTATIONS in capital letters. Sketch your three graphs — small but accurate. Under each sketch write: (1) the name of the graph type, (2) what it shows best, (3) one specific fact it tells us about Mama Wanjiku's canteen.' [Allow 4 minutes. Circulate and check that learners are writing analytical labels, not just drawing shapes.] After 3 minutes: 'Now write one sentence at the bottom of this page — an updated answer to our driving question. Use this sentence starter if you need it: Before today, Mama Wanjiku could not see her data clearly because… but now, using a [graph type], she can see that… and this means she should…' [WAIT 10 seconds before cold-calling.] Cold-call 2 learners to read their updated driving-question answer aloud. [WAIT 10 seconds after each.] Affirm and push: 'Excellent — Zipporah said the pie chart shows ugali takes up 37% of orders. What does that percentage mean in practical terms — if Mama Wanjiku makes 150 portions total tomorrow, how many should be ugali?' [WAIT 10 seconds, cold-call 1 learner.] Close the lesson: 'Your model now has raw data, a frequency table, three measures of	Cumulative Annotated Model Revision — the running model page functions as a personal knowledge artefact that makes the growth of understanding visible over time. Each lesson's addition is a new layer of the same phenomenon, so learners experience statistics not as a list of disconnected procedures (mean, then bar graph, then pie chart) but as an integrated toolkit built around a single motivating problem. The sentence-starter for the driving-question update scaffolds transfer: learners must connect today's graph literacy to Mama Wanjiku's real decision-making context, preventing the model from becoming a decorative drawing exercise.	Teacher checks the analytical labels under each sketch during the circulation phase. Red flags: (1) a learner who labels the line graph as 'best for showing which meal is most popular' has confused graph functions — redirect with: 'Look at your line graph — what is on the horizontal axis? Days of the week. So what question does a line graph answer — a question about amounts, or a question about change over time?'; (2) a learner whose driving-question sentence contains no specific data ('she can now see the information') rather than a data-grounded claim ('she can see that ugali accounts for 37% of all orders, so she needs at least 56 portions per day') needs a follow-up prompt: 'Can you add a number from your graphs to make that sentence stronger?' Collect 6 exercise books at random at the end of the lesson to review model pages and driving-question sentences as written evidence of lesson learning.

		average, and three graph types — all pointing to the same truth that Mama Wanjiku's memory was missing. In Lesson 6 we will learn how to interpret graphs critically — because a graph that is drawn badly or dishonestly can hide the very truth we are trying to reveal.'		

D. TEACHER REFLECTION

1. During the Observe Phase, did all learner pairs correctly calculate pie chart sector angles that summed to 360°? If more than a third made arithmetic errors, should I open Lesson 6 with a brief peer-correction of sector angle calculations before moving to graph interpretation?
2. In the gallery walk, did learners' sticky-note observations focus on visual features ('the ugali bar is the tallest') or on analytical insights ('the pie chart shows ugali is more than a third of all orders')? What does the balance between these two tell me about the depth of graph literacy in this class?
3. When I cold-called learners to compare the bar graph and pie chart, did learners spontaneously connect their answers back to Mama Wanjiku's problem, or did they describe graphs in abstract terms? If the latter, how can I build more explicit phenomenon-linking language into Lesson 6?
4. Were there learners who struggled specifically with using a protractor to measure sector angles — a fine-motor and prior-knowledge gap — and how can I address this without slowing down the whole class in future graph-construction lessons?
5. Looking at the driving-question sentences in the cumulative model pages I collected: are learners' updated answers becoming more specific and data-grounded lesson by lesson, or are they remaining vague? Does the sentence starter scaffold need to be strengthened or removed to promote more independent statistical reasoning?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners constructed three graphs from Mama Wanjiku's canteen frequency data — a bar graph showing meal popularity, a pie chart showing the proportion each meal represents out of 150 total orders, and a line graph showing daily ugali sales across Monday to Friday. In the gallery walk they observed that different graph types made different patterns visible: the bar graph made size comparisons easy, the pie chart communicated proportions (ugali = 37%), and the line graph was the only one to reveal the time-based trend in ugali demand.
What did I learn?	A bar graph uses bar heights to compare frequencies between categories and is ideal for showing which canteen meal is most or least popular. A pie chart uses sector angles — calculated as (frequency ÷ total) × 360° — to show each category as a proportion of the whole, making it clear that ugali accounts for 37% of all orders. A line graph connects plotted data points across a time sequence and is the only graph type that can show whether ugali demand is rising, falling or stable across the school week. The choice of graph type depends on the question being asked, not personal preference.
How does this explain the phenomenon?	Mama Wanjiku's memory-based guesses failed because the human mind cannot accurately track frequency, proportion and time-trend simultaneously across hundreds of transactions. Each graph type makes one of those three dimensions instantly visible: the bar graph would have shown her at a glance that githeri is ordered four times less often than ugali; the pie chart would have shown that nearly two in every five orders are for ugali; and the line graph would have confirmed that ugali demand is consistently high every day of the week, not randomly variable. Together, these three visual tools transform the same frequency table into three different decision-making instruments — exactly the kind of organised, trustworthy information that Mama Wanjiku was missing when she ran out of ugali and was left with unsold githeri.

LESSON 6 (40 minutes): Measures of Central Tendency: Mean and Mode — Finding the 'Typical' Meal for Mama Wanjiku	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners compute the mean and mode of ungrouped data drawn from the canteen phenomenon, interpret what each measure communicates to a decision-maker, and begin to recognise why a third measure of 'typical' may be needed.
Knowledge	Learners understand that the mean is the arithmetic average (sum of all values divided by the number of values) and the mode is the most frequently occurring value in a data set, and that these two measures can yield different answers from the same data.
Skills	Learners calculate the mean and mode of ungrouped discrete data; compare the two measures in context; and annotate their cumulative data model with both statistics.
Attitudes	Learners appreciate that choosing the right statistical measure requires contextual judgement, and demonstrate integrity by recording data honestly when computing measures for their group data sets.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Mama Wanjiku has raw tallies of daily ugali portions sold across five days. In Lesson 5 learners drew bar charts and pie charts of meal-choice frequencies. Now they move from visual patterns to numerical summaries: they compute the mean (58 portions/day) and the mode (60 portions/day) for ugali sales, debate which figure Mama Wanjiku should use when ordering stock, and extend the same tools to Kericho monthly rainfall data — connecting school-canteen decision-making to wider Kenyan agricultural planning. The DQB update plants the seed for the median (Lesson 7).
Safety Notes	No physical hazards. Remind learners to handle printed data sheets with care and return calculators to the charging station after use.

B. LESSON OVERVIEW

Lesson 6 of 8 in the evidence-gathering sequence. Learners use the Mwangaza Secondary School canteen data (ugali portions sold over 5 days: 55, 60, 58, 60, 57) to compute the mean and mode by hand and with a calculator, guided by the teacher through a think-pair-share cycle. They then apply the same procedures to their own group data sets collected in earlier lessons, and briefly examine Kericho monthly rainfall figures as a second Kenyan context. A Driving Question Board (DQB) update closes the lesson: 'We now have mean and mode — is there a third measure of typical?'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles](#) Source: CK-12 🔍 Search ARES	Learners are shown the five-day ugali sales record on the board: Monday 55, Tuesday 60, Wednesday 58, Thursday 60, Friday 57. Before any instruction, each learner silently writes answers to two prompts in their	Display the five data values prominently. Say: 'Look at these five numbers carefully. Without calculating anything yet, write down ONE number you think best represents a typical day for Mama Wanjiku. Explain your reasoning in	Activate-Prior-Knowledge Prediction: surfacing informal notions of 'average' and 'most common' before formal vocabulary is introduced. This creates cognitive dissonance when learners later find that mean ≠	Observable indicator: every learner has a written prediction with a written justification in their book before pair-share begins. Teacher scans for whether learners are guessing randomly or applying informal

for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	exercise books: (a) 'If Mama Wanjiku wants ONE number that represents a typical day's ugali sales, what number would you choose and why?' (b) 'Which day's figure appears most often in the list?' Learners then share predictions with a partner for 45 seconds.	one sentence. You have 90 seconds — go.' [Wait 90 seconds — circulate, read over shoulders, note two or three contrasting predictions.] After 90 seconds: 'Now turn to your partner. Share your number and your reason. You have 45 seconds.' [Wait 45 seconds.] Cold-call 3 pairs with varied answers: 'Amina and Baraka — what did you decide?' 'Chebet and Daud?' 'Esther and Farhan?' Record the three predicted values on the board without judgment. Say: 'We now have three different ideas of what is typical. By the end of this lesson we will have two mathematical tools to check whose prediction was closest — and we will discover why Mama Wanjiku might actually prefer one tool over the other.'	mode, motivating deeper sensemaking.	averaging/frequency reasoning — this informs the depth of scaffolding needed in the Explain phase.
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects (or writes on the board) a worked demonstration of two parallel calculations using the canteen data. Learners follow along, copying each step into their books. Step A — Mean: sum all five values (55+60+58+60+57 = 290), then divide by 5 to get 58. Step B — Mode: identify the value that appears most often — 60 appears twice, all others appear once, so mode = 60. Learners also receive (or copy) a small secondary data table: Kericho average monthly rainfall (mm) for six months — Jan 71, Feb 91, Mar 193, Apr 276, May 216, Jun 38 — and are told they will use this later.	Say: 'We are going to use Mama Wanjiku's data to learn two precise mathematical tools. Watch carefully and copy every step.' Write the heading MEAN on the board. Say: 'Mean = sum of all values divided by how many values there are.' Write the formula: Mean = (Σx) ÷ n. Walk through each arithmetic step aloud, pointing to each value: '55 plus 60 is 115, plus 58 is 173, plus 60 is 233, plus 57 is 290. We have 5 days, so 290 divided by 5 equals... [pause 10 seconds — let learners calculate] ... 58. The mean is 58 ugali portions per day.' Circle 58 on the board. Then write MODE. Say: 'Mode = the value that occurs most frequently. Scan the list. Which number appears more than once?' [Wait 10 seconds.] Cold-call 2 learners. Confirm: 'Sixty appears on	Parallel Worked-Example with Annotation: presenting mean and mode side-by-side from the same data set makes the contrast between the two measures immediately visible, building the conceptual understanding that different statistical tools answer slightly different questions about the same data.	Observable indicator: learners' books show both calculations written out step-by-step with the formula for mean and a circled mode value. Teacher checks 4–5 books during the transition to the next phase for arithmetic accuracy and correct identification of the mode.

		Tuesday and Thursday — twice. Every other value appears only once. So the mode is 60.' Circle 60. Say: 'We now have two different numbers — 58 and 60 — both claiming to describe a typical day. Write this question in your book: Why are they different, and which should Mama Wanjiku use?'		
Explain Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group receives (or has already collected in Lessons 3–4) a small ungrouped data set of 8–10 values — either their own canteen survey responses or a teacher-supplied card with data such as the number of mandazi sold each day, or daily attendance figures at the school gate. Groups compute the mean and mode for their data set. They then answer two discussion questions written on the board: (Q1) 'Why might the mean and mode give different answers for the same data?' (Q2) 'Which measure — mean or mode — would Mama Wanjiku find more useful when deciding how many portions of ugali to prepare tomorrow? Justify your answer.' After group discussion, groups nominate one spokesperson to share their reasoning. The teacher then connects to the Kericho rainfall context: 'A tea farmer in Kericho also uses mean monthly rainfall to plan irrigation. Using the six months of data on the board, calculate the mean monthly rainfall.' (Answer: (71+91+193+276+216+38) ÷ 6 = 885 ÷ 6 = 147.5 mm). Learners identify the mode of the rainfall data (no repeated value — no mode exists), which creates a productive discussion about when mode is and is not a useful	Announce groups and distribute data cards. Say: 'In your group, you have 5 minutes to compute the mean AND the mode for your data set. Show all working. Then discuss the two questions on the board. Nominate a spokesperson.' [Set 5-minute timer. Circulate — listen to at least 3 groups. Prompt struggling groups: 'Start by adding all the values together. How many values do you have?'] After 5 minutes: 'Spokespeople — stand up.' Cold-call 3 groups for Q1 and 3 different groups for Q2. After responses, say: 'Excellent. Now look at the Kericho rainfall table. Calculate the mean monthly rainfall for these six months. Work individually — 2 minutes.' [Wait 2 minutes.] Cold-call 2 learners for the answer. Then ask: 'What is the mode of the Kericho rainfall data?' [Wait 15 seconds.] Take 2 responses. When learners realise there is no mode, say: 'Exactly — no value repeats, so there is no mode. This tells us that mode is most powerful for categorical or discrete data like meal choices, where the same value can repeat frequently. What does this make you wonder about other ways to find the middle of a data set?' [Wait 10 seconds — do not answer; this seeds the DQB update.]	Think-Pair-Share into Whole-Class Synthesis: group computation followed by spokesperson sharing builds collaborative mathematical reasoning. Introducing a contrasting context (Kericho rainfall with no mode) disrupts overgeneralisation and drives learners toward the concept of median before it is formally taught.	Observable indicators: (1) each group's written work shows the mean calculation with correct formula application and a correctly identified or correctly stated 'no mode'; (2) spokesperson explanations for Q2 demonstrate contextual reasoning — e.g., 'Mama Wanjiku should use the mode because 60 is the exact quantity that appeared most often, so preparing 60 portions covers the most common demand' or 'the mean 58 balances out high and low days so she avoids waste.' Teacher notes any group that conflates mean and mode for targeted feedback.

	measure.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Mean and Standard Deviation of Discrete Random Variables (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one new sticky note (or a card on the class DQB section) in response to the prompt: 'We now have mean (58) and mode (60) for Mama Wanjiku's ugali data. But the two numbers are different. Is there a THIRD measure of typical that might help her? What would it look like?' Learners post their cards on the DQB under a new column headed: 'What we still wonder about central tendency.' The class briefly reads three posted cards aloud. The teacher also updates the class's shared phenomenon summary strip: 'Lesson 6 — We can summarise Mama Wanjiku's data with mean = 58 and mode = 60. But which is best — and is there another option?'	Distribute sticky notes or index cards. Say: 'In 2 minutes, write one wondering question on your card. Start with the words: I wonder… Focus on whether mean and mode are enough, or whether Mama Wanjiku needs another number.' [Wait 2 minutes.] 'Post your card on the DQB under the new column I am adding right now.' Add the column label. After all cards are posted, cold-call 3 learners to read their cards aloud. Say: 'Notice that many of you are asking about the number in the MIDDLE of the list — when we order the data from smallest to largest. That idea has a name, and we will explore it in Lesson 7. For now, write this in your book: DQB update — mean and mode answer different questions; a third measure may answer a different question still.'	Driving Question Board as Visible Thinking Routine: public posting of wondering questions makes learners' evolving understanding visible to the whole class, creates intellectual accountability, and generates the motivating question for the next lesson organically from learner thinking rather than teacher telling.	Observable indicator: every learner posts a card with a grammatically complete wondering sentence. Teacher scans for cards that specifically reference ordering the data or finding a middle value — these indicate readiness for Lesson 7 on median. Cards that only restate the mean or mode formula indicate learners who need consolidation.
Model Building Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Mean and Standard Deviation of Discrete Random Variables (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative data model — the annotated diagram or poster they have been building across the sequence to show how raw canteen data is transformed into useful statistics. Today they add two new layers: (a) a 'statistics panel' showing the mean and mode for ugali portions (mean = 58, mode = 60) with the formula for mean written out; (b) an arrow labelled 'What mean tells Mama Wanjiku' pointing to the annotation: 'Average daily demand — useful for budgeting across a week'; and (c) an arrow labelled 'What mode tells Mama Wanjiku' pointing to: 'Most common single-day demand — useful for daily stock decision.' Learners also note on their model: 'Kericho rainfall: mean = 147.5 mm/month; no	Say: 'Take out your cumulative data model. We are going to add today's discoveries.' Display a class version on the board or projector. Walk through each annotation: 'First, add a statistics panel — draw a small box and write: Mean ugali portions = 58; Mode ugali portions = 60. Write the formula: Mean = sum ÷ n.' [Pause 60 seconds for learners to write.] 'Now draw an arrow from mean and write what it tells Mama Wanjiku — average demand across the week, good for weekly budgeting.' [Pause 45 seconds.] 'Draw an arrow from mode and write what it tells her — most common daily demand, good for daily ordering.' [Pause 45 seconds.] 'Now add the Kericho connection in a corner of your	Cumulative Model Revision: each lesson's model update makes the progression from raw data → organised table → graph → numerical summary explicit and visual. Annotating what each statistic means in context (rather than just calculating it) deepens conceptual understanding and connects mathematics to real Kenyan decision-making.	Observable indicator: the model contains both the mean and mode values with correct labels, the mean formula, and distinct contextual interpretations for each measure — not just numbers but sentences explaining their meaning to Mama Wanjiku. Teacher checks 4–5 models during the final minute; learners whose models lack contextual annotations are flagged for a brief check-in at the start of Lesson 7.

	mode — mode only works when values repeat.' At the bottom of the model learners write the open question: '? Median — to be added in Lesson 7.'	model — mean monthly rainfall 147.5 mm, no mode. Same tools, different context.' [Pause 60 seconds.] 'Finally, at the bottom, write a placeholder: Question mark — Median — Lesson 7. Our model is not complete yet.' Cold-call 2 learners to read their model annotations aloud. Close with: 'Every piece we add to this model brings Mama Wanjiku — and a Kericho tea farmer — one step closer to making a decision based on evidence, not memory.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did learners articulate — in their own words — a meaningful difference between what mean and mode each tell Mama Wanjiku, or did they treat the two measures as interchangeable? If most learners struggled to distinguish them contextually, open Lesson 7 with a brief re-examination of the canteen data before introducing the median. (2) How generative were the DQB cards? If fewer than half the class independently referenced ordering the data or finding a middle value, consider adding a brief warm-up in Lesson 7 where learners physically arrange themselves in height order to build the intuition for median. (3) Did the Kericho rainfall extension deepen or distract? If learners found the 'no mode' result confusing rather than illuminating, adjust the pacing in future iterations — the Kericho context may need to be a Lesson 7 opener rather than embedded in Lesson 6's Explain phase. (4) Note which groups produced the strongest contextual justifications for Q2 (mean vs. mode for Mama Wanjiku) — these groups can be deployed as peer-teachers during Lesson 7's consolidation activity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Mama Wanjiku's five-day ugali sales data (55, 60, 58, 60, 57 portions) and Kericho monthly rainfall figures (Jan–Jun) — both presented as raw lists of ungrouped values.
What did I learn?	The mean (sum ÷ number of values = 290 ÷ 5 = 58) and the mode (most frequent value = 60) are two different numerical summaries of the same data set; each answers a slightly different question about what is 'typical,' and mode only works meaningfully when values repeat.
How does this explain the phenomenon?	Mean and mode are both measures of central tendency for ungrouped data. They can give different answers because mean accounts for the size of every value while mode counts only frequency of occurrence. For Mama Wanjiku, mode (60) tells her the most common single-day demand, making it useful for daily stock ordering, while mean (58) smooths out day-to-day variation, making it useful for weekly budgeting. The Kericho rainfall data (mean = 147.5 mm/month, no mode) shows that mode is most powerful for discrete data where repetition occurs — and that a third measure of central tendency (median) may sometimes be needed.

LESSON 7 (80 minutes (2 × 40-minute lessons)): The Middle Value: Median — Finding the Centre of Mama Wanjiku's Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, the learner should be able to determine the median of ungrouped data and interpret what the median reveals about Mama Wanjiku's canteen data — and similar real-world Kenyan datasets — in ways that the mean alone cannot.
Knowledge	The learner will know: (a) the definition of the median as the middle value of an ordered dataset; (b) the rule for finding the median when n is odd (position = (n+1)/2) and when n is even (average of the two middle values); (c) how the median compares to the mean and mode as a measure of central tendency for ungrouped data.
Skills	The learner will be able to: (a) arrange ungrouped data in ascending or descending order; (b) identify the median position using the formula (n+1)/2; (c) calculate the median for both odd-numbered and even-numbered datasets; (d) interpret the median in the context of the canteen phenomenon and other Kenyan community datasets.
Attitudes	The learner will appreciate that the median is a robust, real-world tool for summarising data without being distorted by extreme values, and will display integrity and collaboration while working with peers to organise and interpret data.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	In Lessons 1–3 learners collected Mama Wanjiku's raw lunch-order data and organised it into a frequency distribution table. In Lessons 4–5 they computed the mean (58 ugali portions per day) and identified the mode (60). Now in Lesson 7 they discover the median (57) — the third pillar of central tendency. By comparing all three measures, learners can explain WHY Mama Wanjiku's memory-based guesses failed: her mind tracked the most vivid (mode) day, not the typical middle value. This lesson also prepares learners for Lesson 8, where they will build a complete statistical summary and final visual model to present to 'Mama Wanjiku'.
Safety Notes	No physical safety hazards. Emotional safety note: when using real school data (e.g., learner heights, distances travelled to school), remind learners that data must be handled respectfully and anonymously. No individual's data should be mocked or highlighted negatively.

B. LESSON OVERVIEW

Learners return to Mama Wanjiku's canteen dataset — specifically the daily ugali-portions-sold figures — and discover that the mean (58) they calculated in Lesson 5 can be pulled by unusually high or low days. They are challenged to find a measure that sits at the TRUE MIDDLE of the ordered data. Through a physical ordering activity, a guided calculation routine, and a sensemaking discussion, learners construct the concept of the median for both odd and even datasets. They then compare mean, mode, and median for the canteen data (58, 60, 57) and debate which single number best represents a 'typical' day for Mama Wanjiku's planning. The lesson closes with learners updating the class Driving Question Board and revising their cumulative statistical model of the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Plinko	Learners are presented with a strip of paper showing 7 daily ugali-portions values in RANDOM order:	Distribute the strip of 7 values. Say verbatim: 'Mama Wanjiku wants ONE number she can write on her	Prediction-before-instruction (Eliciting Prior Knowledge): By committing to a written prediction	Observable indicator: Every learner has written a number AND a one-sentence justification on

Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Median (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	54, 60, 57, 62, 55, 58, 60. They are asked to write individually (2 minutes, silent): 'Without doing any calculation, which single number do you think BEST represents a typical day for Mama Wanjiku? Write your choice and explain your reasoning in one sentence.' Learners then share predictions with a shoulder partner (1 minute). Three pairs are cold-called to share — the teacher records all predicted 'typical values' on the board without comment, creating a visible spread of opinions.	whiteboard every morning as her ugali target. Look at these seven days of data. Before we do any maths — which number is the best target, and why? Write your answer silently.' Allow 2 minutes silent thinking. Then say: 'Share with your shoulder partner.' After 1 minute, cold-call exactly 3 pairs: 'Pair at the front-left window — what did you choose?', 'Pair near the door — your choice?', 'Pair at the back-right — what did you decide?' Record each answer on the board in a column headed 'OUR PREDICTIONS'. Do NOT correct or affirm — say: 'Hold that thought. We will come back to check your predictions at the end.'	first, learners activate their intuitive understanding of 'typical value' and create a cognitive anchor. The visible spread of predictions on the board creates productive tension — learners are motivated to find the mathematically precise answer that resolves the disagreement.	their strip within 2 minutes. Teacher scans the room during silent writing — if more than 4 learners are blank after 90 seconds, prompt: 'Just circle the number on your strip that feels most middle to you — then explain why.' Listen to pair-share conversations for key vocabulary: learners who say 'the one in the middle when you sort them' are already close to the median concept; note these learners to anchor Phase 3 discussion.
Observe Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Median (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY 1 — HUMAN NUMBER LINE (15 minutes): Seven volunteers each receive a card with one of the 7 daily values (54, 60, 57, 62, 55, 58, 60). They stand at the front of the class in the random order they received the cards. The class is asked: 'Can you identify the middle value NOW?' (Likely answer: No — it is disordered.) Learners then direct the 7 volunteers to rearrange themselves in ascending order. Once ordered (54, 55, 57, 58, 60, 60, 62), the class identifies the person standing in the middle position. That volunteer holds up their card: 58. The teacher asks: 'How many people are standing to the LEFT of the middle person? How many to the RIGHT?' (Answer: 3 each side.) Learners record: 'The MEDIAN is the middle value of ORDERED data.' ACTIVITY 2 — EVEN DATASET CHALLENGE (10 minutes): Teacher adds an 8th card: 56 (representing a new Monday	For Activity 1: Select 7 volunteers and hand out value cards face-down. Say: 'Do NOT show your card yet.' Once standing randomly: 'Class — can anyone tell me who has the middle value? ... [WAIT 10 seconds] ... Why not? Turn and tell your partner in 10 seconds.' Cold-call 2 pairs for responses. Then: 'Class — give instructions to arrange our volunteers from smallest to largest. Call out the moves.' Facilitate physical reordering. Once ordered, ask: 'Count the people to the left of Position 4 — how many? [WAIT 10 seconds] To the right? [WAIT 10 seconds] What do we call this middle value?' Write on board: MEDIAN = middle value of ORDERED data. For the position formula, write: 'Position of median = (n+1)/2. With n=7: (7+1)/2 = 4th value.' For Activity 2: Introduce 8th card. Ask: 'We now have 8 values — an even number. Where is the middle? [WAIT 15 seconds] ... Discuss with your partner.' Cold-	Embodied Cognition (Human Number Line): Making learners physically embody data values leverages kinesthetic and spatial reasoning. The act of re-ordering peers makes the abstraction of 'ordered data' concrete and memorable. The 'even number' challenge creates deliberate cognitive conflict that motivates the averaging rule — learners feel the problem before they learn the solution.	Observable indicators: (1) During the human number line, all seated learners should be able to count 3 values on each side of the median — if learners are unsure, the ordering has not been grasped; (2) In Activity 2, listen for learners who spontaneously say 'we average the two middle ones' — this signals strong prior understanding; (3) In Activity 3 pair work, circulate and check: are learners writing the data in order BEFORE identifying the median? Flag any pair that writes '49' (unordered middle) as the median — this is the key misconception to address in Phase 3.

	reading). Volunteers re-order to: 54, 55, 56, 57, 58, 60, 60, 62. Teacher asks: 'Where is the ONE middle person now?' Learners discover there are TWO middle people (57 and 58). Teacher guides: 'When n is even, we take the AVERAGE of the two middle values.' Learners calculate: (57+58)/2 = 57.5. Learners record both rules in their exercise books with a worked example each. ACTIVITY 3 — CANTEEN DATASET (5 minutes): Learners work in pairs to find the median of a 5-day maize-portions dataset: 44, 51, 49, 53, 47. They order the data and identify the median independently.	call 3 pairs. Guide to average formula. Write on board: 'When n is EVEN: Median = (value at n/2 position + value at n/2 + 1 position) ÷ 2.' For Activity 3: Say: 'In your pairs — order the 5 maize values and find the median. You have 4 minutes.' Circulate and note pairs who skip the ordering step (common error).		
Explain Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Median (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — FORMALISING THE RULES (10 minutes): Learners copy and complete a structured note frame in their books: 'MEDIAN of ungrouped data: Step 1 — ___ the data in ___ order. Step 2 — Count the number of values (n). Step 3 — If n is ODD, median = value at position ___. If n is EVEN, median = ___.' They then apply the rules independently to two new Kenyan-context problems: (i) Heights (in cm) of 9 Form 4 athletes from Rift Valley selected for the national team: 172, 168, 175, 180, 165, 178, 171, 169, 174 — find the median height. (ii) Daily chapati sales at Mama Wanjiku's canteen over 6 days: 38, 45, 41, 50, 36, 43 — find the median chapati sales. PART B — THE BIG COMPARISON (10 minutes): Learners are given a reference card showing all three measures for the canteen ugali data: Mean = 58, Mode = 60, Median = 57. In groups of 4, learners discuss: 'Mama Wanjiku	For Part A note frame: Distribute printed frames or have learners copy from board. Say: 'Fill in the blanks using what we just discovered with our human number line.' Allow 3 minutes silent work. Cold-call 2 learners to read their completed frames aloud — correct any errors publicly and precisely. For the two practice problems, say: 'Work independently — no partner help yet — for 5 minutes. Show ALL steps: write the ordered list, write n, write the position formula, circle the median.' Circulate and look for the ordering step — stamp or tick books where ordering is shown explicitly. For Part B, distribute the Mean/Mode/Median reference card. Say: 'In your groups of four, I want you to imagine Mama Wanjiku adds the sports-day data point: 92 portions. Discuss: which measure — mean, mode, or median — changes the most? You have 4 minutes. Write two sentences of reasoning.' After 4	Concept Mapping through Comparison (Three Measures Side-by-Side): Placing mean, mode, and median together forces learners to build a relational understanding rather than treating each as an isolated calculation. The sports-day outlier scenario is a deliberate 'stress test' of each measure — learners discover the median's robustness by reasoning, not by being told. This connects directly back to the phenomenon: Mama Wanjiku's memory was dominated by memorable extreme days (like sports day), analogous to the mean being pulled by outliers.	Observable indicators: (1) Note-frame completion — learners must write '(n+1)/2' in the odd-number blank; accept equivalent correct phrasing. (2) For the athlete heights problem (n=9, ordered: 165,168,169,171,172,174,175,178,180 → median = 172), circulate and check that learners show the ordered list before circling position 5. (3) For chapati problem (n=6, ordered: 36,38,41,43,45,50 → median = (41+43)/2 = 42), listen for learners who average 41 and 43 correctly. (4) For Part B group task, the target reasoning is: 'Mean increases significantly because 92 is far above average; median barely changes because it only depends on the middle position.' Any group that identifies this mean-sensitivity is demonstrating mastery-level understanding.

	had one very unusual day — a school sports day — when she sold 92 ugali portions. If we add this day to the dataset, which measure changes the most: mean, mode, or median? Why does this matter for her daily planning?' Groups record their reasoning in two sentences. Three groups present — teacher facilitates whole-class comparison.	minutes, cold-call 3 groups: 'Group near the window — what did you decide and why? [WAIT 10 seconds before prompting]'. Synthesise on board: 'The MEAN is sensitive to extreme values (outliers). The MEDIAN is resistant — it barely moves. This is why statisticians say the median is a ROBUST measure.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Median (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes of different colours: ORANGE = 'Something I can now EXPLAIN about Mama Wanjiku's problem that I couldn't explain before Lesson 7' and GREEN = 'A new question I now have about statistics or data.' Learners write independently for 2 minutes, then post their notes on the class Driving Question Board under the headings 'NEW UNDERSTANDING' and 'NEW QUESTIONS'. The teacher reads 3–4 notes aloud and links them explicitly to the driving question: 'How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?' The teacher highlights that learners now have THREE statistical tools (mean, mode, median) and asks: 'What is STILL missing from our toolkit before we can give Mama Wanjiku a complete recommendation?' (Leading to Lesson 8: full summary, graph interpretation, and final presentation.)	Distribute sticky notes. Say: 'ORANGE note — write one thing you can now explain about Mama Wanjiku's canteen problem because of what we learned today about the median. GREEN note — write one new question our lesson has raised for you.' Allow exactly 2 minutes silent writing — enforce silence. Then: 'Post your notes on the DQB.' Read 3 orange and 2 green notes aloud. For each orange note, affirm and extend: e.g., 'Wanjiru writes: I can now explain that the median is not affected by the sports-day spike — excellent! This is exactly why Mama Wanjiku should use the median as her daily target, not the mean.' For green notes, say: 'These questions are exactly what drives a statistician. We will answer the question about graphs and charts in our final lesson.' Close by writing on the board: 'Lesson 7 unlocks: MEDIAN. We now have Mean ✓ Mode ✓ Median ✓. Lesson 8: Putting it all together for Mama Wanjiku.'	Driving Question Board (DQB) Curation: The DQB is a living artefact of the class's growing understanding. Colour-coded notes make PROGRESS (orange) and CURIOSITY (green) visible simultaneously. Reading notes aloud validates learner thinking and models how scientists track what they know versus what they still need to investigate — mirroring the NGSS practice of 'Obtaining, Evaluating, and Communicating Information'.	Observable indicators: (1) Every learner posts both colours — absence of a sticky note signals disengagement or confusion; approach those learners immediately. (2) Orange notes should reference the canteen phenomenon specifically — generic answers like 'I learned about median' are a flag; push with 'How does this help Mama Wanjiku?' (3) Green notes that ask about graphs, outliers, or 'what if the data had more values' signal readiness for Lesson 8. (4) Green notes that re-ask questions already answered in prior lessons signal a need for review — note these learners for targeted follow-up.
Model Building Phase 📹 **VIDEO:**	Learners open their cumulative Statistical Model diagrams — a visual they have been building since Lesson 1, showing how raw	Say: 'Open your Statistical Model diagrams from Lesson 1. You are going to add today's discovery.' Display the teacher's model on the	Cumulative Model Revision (NGSS Science and Engineering Practice 2 — Developing and Using Models): Each lesson adds	Observable indicators: (1) All three measures correctly labelled on the model with accurate values (Mean=58, Mode=60, Median=57).

[Video: Plinko Probability](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Median (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	canteen data transforms into organised information. In Lesson 5 they added the MEAN arrow. In Lesson 6 they added the MODE arrow. Today they add a third arrow labelled MEDIAN and annotate it with: (a) the formula (n+1)/2 for odd n; (b) the value for Mama Wanjiku's ugali data (Median = 57); (c) a comparison box: 'Mean = 58, Mode = 60, Median = 57 — all close together, suggesting the data is fairly symmetric with no extreme skew.' Learners also add a 'ROBUSTNESS NOTE' box: 'Median is the best measure for planning when extreme values (e.g., sports-day spikes) might distort the mean.' Learners then write a two-sentence 'Advice to Mama Wanjiku' at the bottom of their model: 'Based on the median, Mama Wanjiku should prepare approximately ___ portions of ugali each day. This is better than using the mean because ___.'	board (or projector). Point to the existing mean and mode arrows: 'You added MEAN in Lesson 5 and MODE in Lesson 6. Today you add MEDIAN.' Model the annotation live on the board — draw the arrow, write the formula, write 57, draw the comparison box. Say: 'Now copy and add to your own diagram — you have 5 minutes.' Circulate and check that all three measures (mean, mode, median) are present and correctly labelled. After 5 minutes, say: 'At the bottom of your model, write your two-sentence Advice to Mama Wanjiku. Use your own words. You have 3 minutes.' Cold-call 3 learners to read their advice aloud. After each reading, ask the class: 'Do you agree with this advice? Thumbs up if yes, thumbs sideways if partly, thumbs down if you disagree.' Address any thumbs-down responses before closing.	one layer to the model, making the transformation from raw data to decision-making tool visually explicit. By writing 'Advice to Mama Wanjiku', learners practise the NGSS practice of 'Constructing Explanations' — translating statistical findings into actionable real-world recommendations. The comparison box (Mean/Mode/Median side-by-side) scaffolds the understanding that measures of central tendency must be interpreted together, not in isolation.	(2) The ROBUSTNESS NOTE must explicitly mention that the median is resistant to outliers — vague statements like 'median is good' are insufficient; prompt: 'What happens to the median if we add the sports-day value of 92?' (3) The 'Advice to Mama Wanjiku' sentence must include a specific number (57 or approximately 57) AND a comparative justification. Target response example: 'Based on the median, Mama Wanjiku should prepare approximately 57 portions of ugali each day. This is better than using the mean because if one unusually busy day like sports day is included, the mean rises but the median stays stable, giving a more reliable daily target.' Collect 5 model diagrams as exit evidence for teacher review before Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the Human Number Line activity make the ordering step concrete enough — or did learners still skip it when working independently in Phase 3? Note which pairs needed prompting to order data first. (2) During the Part B group discussion (sports-day outlier), did groups genuinely reason about WHY the mean is sensitive to extreme values, or did they simply repeat the conclusion? If reasoning was shallow, plan a brief review at the start of Lesson 8 using a new Kenyan context (e.g., monthly rainfall in Kisumu). (3) Review the 5 collected model diagrams: are the Advice to Mama Wanjiku sentences specific and justified? Learners who write only 'she should use the median' without a number or a comparative reason will need targeted scaffolding in Lesson 8. (4) Scan the green sticky notes on the DQB: if multiple learners are asking about what happens with very large datasets or grouped data, consider a brief preview bridge at the start of Lesson 8. (5) Was the 80-minute double period paced well? The Human Number Line can expand if learners are highly engaged — protect at least 15 minutes for Model Building (Phase 5), as this is the cumulative artefact for the whole unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically ordered 7 daily ugali-portion values from Mama Wanjiku's canteen data using a Human Number Line activity, and identified the middle value (58). They then observed what happens when an 8th value (56) is added, creating an even-numbered dataset with two middle values (57 and 58), requiring them to calculate the average (57.5). They observed that the

	median barely changed when the sports-day outlier of 92 portions was added to the dataset.
What did I learn?	Learners learned that the MEDIAN is the middle value of a dataset that has been arranged in ascending (or descending) order. For an odd number of values (n), the median is at position (n+1)/2. For an even number of values, the median is the average of the two middle values. The median is a ROBUST measure — unlike the mean, it is not significantly affected by extreme values (outliers) such as an unusually busy sports-day at the canteen. For Mama Wanjiku's 7-day ugali data, the median is 57 portions per day.
How does this explain the phenomenon?	Learners explained that Mama Wanjiku's memory-based guesses failed partly because her memory was anchored to the most vivid or extreme days (like sports day with 92 portions), which is analogous to the mean being pulled upward by outliers. The median gives a more reliable daily planning target (57 portions) because it reflects the true MIDDLE of her typical trading days, unaffected by exceptional events. Learners updated their cumulative Statistical Models with the median arrow and wrote personalised two-sentence recommendations to Mama Wanjiku, integrating mean, mode, and median into a coherent statistical argument.

LESSON 8 (40 minutes): The Full Picture — Final Explanation: Giving Mama Wanjiku a Complete Statistical Report	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all statistical tools — frequency distribution tables, mean, mode, median, bar graphs, line graphs and pie charts — into a single coherent written report that directly answers Mama Wanjiku's canteen planning problem, completing the storyline and demonstrating full competency in Statistics 1.
Knowledge	– Frequency distribution tables organise raw data into readable counts that reveal patterns invisible in an unordered list. – Mean, mode and median are three complementary measures of central tendency that each answer a slightly different planning question for ungrouped data. – Bar graphs, line graphs and pie charts each represent different aspects of the same data set and must be chosen based on the question being answered.
Skills	– Construct a complete statistical report by integrating frequency tables, all three measures of central tendency and at least two graph types for a real data set. – Interpret statistical outputs (numerical and graphical) to make a justified, evidence-based recommendation to a real decision-maker (Mama Wanjiku). – Compare an early naive model (Lesson 1 'memory guess') with a final evidence-based statistical model and articulate what changed and why.
Attitudes	– Appreciate that honest, organised data — not memory or guesswork — is the foundation of fair and effective decision-making in Kenyan community life. – Value the collaborative process of statistical inquiry as a civic skill that empowers ordinary people, including school canteen managers, farmers, and community leaders, to solve real problems.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	This is the culminating lesson of the Statistics 1 unit. Learners deploy every tool built across Lessons 1–7 — tally sheets, frequency distribution tables, mean, mode, median, bar graphs, line graphs, and pie charts — to produce a final integrated statistical report for Mama Wanjiku. By comparing their Lesson 1 'memory-based guess' model with their fully evidenced final model, learners see exactly how raw data is transformed into decision-making power. The Driving Question Board is formally closed with agreed class answers, and learners articulate how the same statistical toolkit can scale beyond the canteen to transport, health, agriculture, and environment across Kenya.
Safety Notes	No specific hazards. Ensure learners handle shared rulers and graph paper carefully. If using digital devices for graph drawing, apply school ICT safety guidelines.

B. LESSON OVERVIEW

Lesson 8 is the final explanation lesson of the Statistics 1 sub-strand and the resolution of the eight-lesson Canteen Queue Puzzle storyline. Across the previous seven lessons, learners have built a growing toolkit: they designed tally sheets (Lesson 2), constructed frequency distribution tables (Lessons 1 and 4), computed mean and mode (Lesson 6), found the median (Lesson 7), and drew bar graphs, line graphs and pie charts (Lesson 5). Today, all of those tools converge. Working in their established groups, learners produce a short but complete statistical report addressed directly to Mama Wanjiku — a document that contains a frequency distribution table, all three measures of central tendency with interpretation, at least two graph types, and a clear, justified recommendation about how many portions of each meal she should prepare on a typical school day.

The lesson opens with a Predict Phase in which learners revisit the very first artefact of the unit — the raw, unorganised list of 120 lunch orders from Lesson 1 — and compare it with their fully developed group data model. This side-by-side comparison anchors the lesson's core cognitive move: articulating the transformation from noise to knowledge. In the Observe Phase, groups lay out all their work from Lessons 1–7 and conduct a 'statistical audit', checking that every component is accurate and complete before writing the report. In the Explain Phase, groups present their final reports, receive peer feedback using a structured sentence frame, and refine their recommendations. The teacher cold-calls individual learners to justify specific statistical choices ('Why did your group choose the mode rather than the mean to recommend ugali portions?'), providing 15-second wait time before calling on a named learner.

The lesson closes with a formal Driving Question Board completion ceremony: the class agrees on written answers to each question posted across the eight lessons and seals the board. In the Model Building Phase, learners draw their final cumulative model — a one-page annotated diagram showing the full journey from Mama Wanjiku's memory guess to a statistically grounded canteen plan — and write a 'transfer challenge' sentence explaining how the same statistical process could help a maize farmer in the Rift Valley, a matatu operator in Nairobi, or a public health officer tracking malaria cases near Lake Victoria. This transfer task directly answers the driving question at its broadest scale.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner individually receives (or views on screen) two documents side by side: (A) the original raw data list from Lesson 1 — 120 unorganised meal-choice entries — and (B) their group's fully completed data model containing the frequency table, three measures of central tendency, and graphs. Learners write a 3-sentence response in their exercise books answering: 'What could Mama Wanjiku actually do with document A? What can she do with document B? What made the difference?' After writing, learners share with a shoulder partner for 90 seconds before the teacher takes whole-class cold calls.	Distribute or display the two documents. Say: 'Before we write Mama Wanjiku's final report, I want us to go back to the very beginning. Look at document A — the raw list from our first lesson. Look at document B — everything your group has built. In your book, write three sentences: What can Mama Wanjiku do with A? What can she do with B? What made the difference? You have 3 minutes — go.' [Allow 3 minutes of silent individual writing. Then say:] 'Share your three sentences with your shoulder partner — 90 seconds.' [After 90 seconds:] 'I am going to cold-call four people. Remember: I will wait 15 seconds before I call a name, so everyone should be ready.' Pause 15 seconds, then cold-call Learner 1: 'Achieng, what can Mama Wanjiku do with the raw list?' After response, cold-call Learner 2: 'Kamau, what can she do with the	Contrastive Analysis (Before/After Comparison): Placing the raw unorganised data next to the fully developed statistical model forces learners to name, in their own words, the cognitive and procedural work done across all eight lessons. This strategy activates prior knowledge from all seven previous lessons simultaneously and creates a felt sense of progress that motivates the final synthesis task.	Teacher scans exercise books during the 3-minute writing period — look for learners who can name at least one specific statistical tool (e.g., 'the frequency table showed us which meal is most popular') rather than giving vague answers. During cold calls, listen for the key idea: 'organising/calculating/visualising the data' as the transformative step. Learners who say only 'the graph looks nicer' need a follow-up probe: 'But what decision can Mama Wanjiku make from the graph that she could not make from the raw list?'

		completed model?' Cold-call Learner 3: 'Fatuma, what made the difference?' Cold-call Learner 4: 'Omondi, would you add or change anything Fatuma said?' Write key phrases from learner responses on the board — these will be revisited when closing the DQB.		
Observe Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Ungrouped Data to Find the Mean (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups (same groups as Lessons 1–7) spread all their accumulated work on their desks: tally sheets, frequency distribution tables, calculations of mean, mode and median, and the three graph types. They conduct a 'Statistical Audit' using a five-item checklist (written on the board by the teacher): (1) Frequency distribution table with correct tallies and totals summing to 120; (2) Mean calculated correctly — 290 ÷ 5 = 58; (3) Mode identified correctly — 60; (4) Median identified correctly — 57; (5) At least two graph types drawn with correct labels, titles, and axes. Groups tick each item, correct any errors, and assign each group member one section of the final written report to draft (the report has four sections: Introduction, Data Table, Measures of Central Tendency, Graphs and Recommendation).	Write the five-item Statistical Audit checklist on the board and say: 'This is your quality check before you write the final report. Every item must be ticked — no blanks allowed. If you find an error, fix it now. You have 7 minutes.' Circulate actively — visit every group at least once. At 4 minutes, give a verbal warning: 'Three minutes remaining for the audit — finalise your corrections.' At 7 minutes, call attention: 'Hands up if your group found at least one error during the audit.' Acknowledge responses without judgment: 'That is exactly what audits are for — catching errors before they reach the decision-maker.' Then say: 'Now assign report sections. I want to see every group member holding a pen and writing their section in the next 10 minutes. The report is addressed to Mama Wanjiku — write it as if she will actually read it, because she will.' [Teacher continues circulating, prompting groups where one learner is dominating the writing.]	Statistical Audit Protocol: A structured self-assessment checklist that requires learners to verify numerical accuracy and completeness of all statistical outputs before synthesis. This mirrors authentic scientific and professional practice (data verification before reporting) and reinforces the lesson's core message that statistical work must be accurate to be useful — Mama Wanjiku will make real purchasing decisions based on this report.	Teacher checks that frequency totals sum to 120 in every group's table — a total ≠ 120 signals a tally or arithmetic error that must be corrected before the report is written. Teacher checks that all three measures of central tendency are present and correctly labelled. Listen for group discussions about which graph type to include — groups that can articulate 'we are using the bar chart to compare meals and the pie chart to show proportions' demonstrate deeper understanding than groups that choose graphs arbitrarily.
Explain Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos	Three groups present their final reports to the class (selected by the teacher to represent a range of approaches). Each presentation lasts 3 minutes maximum: one learner presents the data table, one presents the measures of central tendency, one presents the graphs, and one delivers the	Before presentations begin, write the peer-feedback sentence frame on the board. Say: 'Three groups will present. While they present, every other group writes feedback using this sentence frame. I will cold-call two people from the audience after each presentation — not the presenter's group — to	Gallery-Style Structured Presentations with Peer Feedback Frames: Requiring presenters to address a real audience (Mama Wanjiku, represented by the class) and receive structured feedback develops both statistical communication skills and metacognitive awareness. The	Teacher listens specifically for whether recommendations are evidence-based: 'Mama Wanjiku should prepare 60 portions of ugali because the mode is 60' is acceptable; 'Mama Wanjiku should prepare about 58 portions' using the mean is also acceptable; the strongest responses will

📖 **HTML:** [Ungrouped Data to Find the Mean (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	recommendation to Mama Wanjiku. After each presentation, the class gives structured peer feedback using the sentence frame written on the board: 'Your report helped Mama Wanjiku by ___. One thing you could add or clarify is ___.' Presenting groups take notes on the feedback for a final 2-minute revision of their reports at the end of the phase.	share their feedback aloud. Wait time is 15 seconds, so be ready.' [After Group 1 presents:] Pause 15 seconds, then cold-call: 'Njeri, what is your feedback for Group 1?' Then: 'Wafula, do you agree, or do you have a different view?' After all three presentations, ask the whole class: 'Every group used mean, mode and median. I want to ask one deeper question — why did we need all three? Why wasn't the mode alone enough for Mama Wanjiku?' Pause 15 seconds. Cold-call: 'Chebet, what do you think?' Then: 'Abdi, can you add to that?' Anticipated answer to draw out: 'The mode tells us the most common amount, but the mean balances out all the days, and the median is not pulled up by one unusually busy day — together they give Mama Wanjiku a fuller picture.' Write this synthesis on the board verbatim as it emerges from learners.	sentence frame ('Your report helped Mama Wanjiku by…') keeps feedback anchored to the phenomenon rather than drifting into generic praise, ensuring that every comment made in the room reinforces why statistics matter for real Kenyan decision-makers.	acknowledge all three measures and recommend a range (e.g., 57–60 portions) with explicit reasoning. Peer feedback quality also reveals understanding — feedback that references specific numbers ('your mode was listed as 58 but from the table it should be 60') shows higher-order engagement than generic comments.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Ungrouped Data to Find the Mean (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals the full Driving Question Board — displayed at the front of the room and containing every question that has been added across Lessons 1–7. These include questions such as: 'Why does the mean change when we add a new data point?', 'Which graph is best for comparing meals?', 'Why does ordering matter for median but not for mode?', and 'Can these tools help beyond the canteen?' Learners, in pairs, are assigned one question card each and write a 2-sentence answer on a sticky note in a different colour (green = answered, replacing the original question card). The class then reviews the completed board together, with the teacher reading each green	Gesture to the DQB and say: 'This board has held our questions since Lesson 1. Today we close it — every question gets a green answer. Work with your partner — you have one question card. Write your answer in two sentences on a green sticky note. You have 4 minutes.' [Circulate to support pairs who are stuck — prompt with 'Look at your report. Which part of your statistical work answers this question?'] After 4 minutes, lead the gallery walk: read each green answer aloud. For contested answers, invite a show of hands and a brief 30-second discussion. Finally, say: 'Now for the big one — our driving question. I will read it aloud. Then I will give you 15 seconds of thinking time, and I	Driving Question Board Closure Protocol: Physically replacing question cards with answer cards — in a different colour — makes the growth of understanding tangible and visible. Learners experience the satisfying intellectual closure of a completed inquiry cycle, which consolidates the unit's narrative arc. Answering the driving question collectively as a final oral synthesis ensures that the big idea ('raw data + statistical tools = decision-making power') is explicitly stated in learners' own language before the lesson ends.	Teacher reads every green answer before accepting it — look for answers that are too vague ('we learned about statistics') and prompt revision: 'Can you name one specific tool and one specific decision it supports?' The driving question answer is the most important formative indicator: a strong answer will name at least two statistical tools (e.g., frequency table + bar chart, or mean + pie chart) AND connect them to a specific decision outcome for Mama Wanjiku or another Kenyan decision-maker. Weak answers that cannot name tools signal a learner who needs additional consolidation support.

	answer aloud and asking for a show of hands: 'Do we agree this question is fully answered?' The driving question itself — 'How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?' — is answered last, collectively, as a class oral statement that the teacher scribes on the board.	want one volunteer to give us the class answer in two or three sentences.' Read the driving question slowly. Pause 15 seconds. Take the volunteer's response, then say: 'Let me write that on the board as our official class answer.' Scribe the answer, then say: 'That answer — that is what eight lessons of statistics looks like. Mama Wanjiku can now walk into the canteen on Monday morning with a plan based on evidence, not memory.'		
Model Building Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner independently draws their Final Cumulative Model — a one-page annotated diagram in their exercise book. The model must show the full journey from left to right: (1) Mama Wanjiku's memory guess (raw data, no organisation) → (2) Tally sheet and frequency distribution table → (3) Mean = 58, Mode = 60, Median = 57 → (4) Bar graph / pie chart / line graph → (5) Evidence-based recommendation to Mama Wanjiku. Arrows connect each stage with a one-word label describing what the statistical tool does (e.g., 'organises', 'summarises', 'visualises', 'decides'). Below the model, each learner writes one 'Transfer Sentence': 'The same statistical process that helped Mama Wanjiku could also help a ___ in ___ [Kenyan place] to ___, by using a ___ to organise the data and a ___ to show the pattern.' Example sentence: 'The same statistical process that helped Mama Wanjiku could also help a maize farmer in the Rift Valley to plan how much fertiliser to buy, by using a frequency distribution table to organise weekly rainfall records	Say: 'For the final 8 minutes, you are working alone. Draw your Final Model — the complete journey from raw data to decision. Label every stage with an arrow and a one-word verb. Then write your Transfer Sentence at the bottom. The transfer sentence must name a real Kenyan place and a real decision-maker — not Mama Wanjiku. Think about farmers in Kericho, matatu operators in Nairobi, nurses tracking malaria cases near Lake Victoria, or coaches recording athletes' sprint times at Nyayo Stadium.' [Allow 8 minutes of silent individual work. Circulate and read transfer sentences — quietly affirm strong ones: 'Excellent — you have named the tool AND the decision.'] With 2 minutes remaining, say: 'I will cold-call three people to read their transfer sentence aloud — this is not for marks, it is to inspire the whole class.' Pause 15 seconds, then cold-call three learners from different parts of the room. After each reading, say: 'Thank you — that shows that statistics is not just a canteen tool. It is a community tool.' Close the lesson: 'Every one of you has now	Cumulative Model Revision with Transfer Sentence: The final model revision requires learners to represent the entire unit's conceptual arc in a single visual — the highest-order synthesis task of the lesson. The mandatory Transfer Sentence operationalises the KICD core competency of 'applying mathematics in real life situations' and directly closes the driving question at its broadest scale, linking school-level statistics to community and national decision-making across Kenya.	Teacher collects or photographs all final models for summative assessment. Evaluate against three criteria: (1) All five stages are present and correctly labelled — indicates completeness of conceptual understanding; (2) Arrow labels use process verbs ('organises', 'summarises', 'visualises', 'decides') rather than just tool names — indicates understanding of what each tool does, not just what it is called; (3) The Transfer Sentence names a specific Kenyan context, a specific decision-maker, a specific data type, and at least one named statistical tool — indicates ability to transfer statistical thinking beyond the anchoring phenomenon. Learners who cannot write a transfer sentence independently should be given a follow-up scaffolded task in the next lesson or during personal study time.

	and a line graph to show the rainfall trend.'	built a complete statistical toolkit. Mama Wanjiku's problem is solved. And now you can see how to solve the next one.'		

D. TEACHER REFLECTION

1. During the Statistical Audit, which groups found errors in their frequency tables or measures of central tendency — and what type of errors were most common (arithmetic, labelling, wrong formula)? What does this pattern tell me about which concepts need consolidation or re-teaching before the end-of-strand assessment?
2. When I cold-called learners to justify why all three measures of central tendency are needed, did most learners articulate a meaningful distinction between mean, mode and median — or did they conflate them? Do I need to design a short consolidation activity for the Probability 1 lessons that briefly revisits this?
3. How was the quality of the peer feedback during presentations? Did learners use the sentence frame to make specific, evidence-referenced comments, or did they give generic praise? What does this tell me about my learners' ability to engage in mathematical communication — a core KICD competency?
4. Looking at the completed Driving Question Board, which questions remained partially answered or generated disagreement during the closure ceremony? Are there conceptual gaps that I need to address through the Statistics 1 assessment task or the project component?
5. How varied and ambitious were the Transfer Sentences learners wrote in the Model Building Phase? Did learners connect statistics only to food contexts (showing limited transfer), or did they reach for health, transport, agriculture and environment contexts? What does this tell me about how well this eight-lesson storyline succeeded in building a generalisable statistical mindset — and how can I strengthen this transfer dimension if I teach this unit again?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed the complete contrast between the original raw, unorganised list of 120 canteen meal orders from Lesson 1 and their fully built statistical model from Lessons 1–7, seeing in a single side-by-side comparison exactly how each tool (tally sheet → frequency table → mean/mode/median → graphs) added a layer of organisation, summarisation or visualisation to transform invisible data into an actionable recommendation for Mama Wanjiku.
What did I learn?	Learners consolidated that frequency distribution tables, mean, mode, median, bar graphs, line graphs and pie charts are not isolated skills but a coherent, sequential statistical process: raw data must first be organised (frequency table), then numerically summarised (measures of central tendency), then visually represented (graphs), before any evidence-based decision can be made. They also learned that the same process scales beyond Mama Wanjiku's canteen to any real Kenyan decision-maker — a maize farmer in the Rift Valley, a matatu operator in Nairobi, a nurse tracking disease cases near Lake Victoria — who needs to transform raw data into knowledge.
How does this explain the phenomenon?	The anchoring phenomenon is now fully explained: Mama Wanjiku's memory-based guesses failed because raw, unorganised data is essentially invisible information — the human memory cannot reliably detect statistical patterns across 120 data points without systematic tools. A frequency distribution table made the pattern visible; measures of central tendency gave her a specific, defensible planning target (prepare approximately 57–60 portions of ugali per day); and graphs allowed her to communicate the finding to others at a glance. The Canteen Queue Puzzle is solved — and the statistical toolkit that solved it is the same one that can be applied to any data-rich problem in Kenyan community life.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.